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Tumors mimic the niche to inhibit neighboring stem cell differentiation

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eLife Assessment

This study presents results supporting a model that tumorous germline stem cells (GSCs) in the Drosophila ovary mimic the stem cell niche and inhibit the differentiation of neighboring cells. The **valuable** findings show that GSC tumors often contain non-mutant cells whose differentiation is suppressed by the GSC tumorous cells. However, the evidence showing that the GSC tumors produce BMP ligands to suppress differentiation of non-mutant cells is **incomplete**. It could be strengthened by the use of sensitive RNA in situ hybridization approaches.

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Abstract

Although it is well-established that stem cells maintain tissue homeostasis while tumors disrupt it, the mechanisms by which tumors influence the development of nearby stem cells remain poorly understood. Using *Drosophila* ovaries as a model system, here we discovered that *bam* or *bgcn* mutant germline tumors inhibit the differentiation of neighboring wild-type germline stem cells (GSCs). Mechanistically, these tumor cells mimic the stem cell niche by secreting the BMP ligands Dpp and Gbb, but at reduced levels, resulting in moderate BMP signaling activation in adjacent GSCs. Such BMP signaling activation is sufficient to repress *bam* transcription, thereby blocking GSC differentiation. To our knowledge, this is the first example that tumors can functionally mimic a stem cell niche to inhibit the differentiation of neighboring wild-type stem cells. Similar regulatory paradigms may operate in mammalian tissues, including humans, during tumorigenesis.

Introduction

The homeostasis of many tissues in our bodies is maintained by adult stem cells, but this balance can be disrupted by tumor cells. What occurs when tumorigenesis intersects with stem cell development? To address this question, a mosaic-analysis model system is essential, where wild-type stem cells develop alongside tumor cells. *Drosophila* offers an exceptional model for such studies, as it allows for the efficient generation of mosaic clones through various established methods (Germani et al., 2018; Pastor-Pareja and Xu, 2013).

In *Drosophila* ovaries, germline stem cells (GSCs) play a crucial role in sustaining normal oogenesis and maintaining fertility (Fuller and Spradling, 2007; Lin, 1997). These GSCs reside in a specialized microenvironment known as the stem cell niche (hereafter referred to as niche) (Xie and Spradling, 2000). Typically, a GSC undergoes asymmetric division, generating two distinct daughter cells: one remains in the niche to self-renew as a GSC, while the other, called a cystoblast, exits the niche and initiates differentiation. During the differentiation process, each cystoblast performs exactly four rounds of mitotic division with incomplete cytokinesis to produce16 interconnected cystocytes, forming a germline cyst. In each germline cyst, only one germ cell is destined to become the oocyte, while the remaining 15 differentiate into nurse cells that support the development of the oocyte (***Figure 1A***) (Fuller and Spradling, 2007; Lin, 1997). The principal niche signals are bone morphogenetic protein (BMP) ligands, including Decapentaplegic (Dpp) and Glass bottom boat (Gbb), which are secreted predominantly by cap cells, the major somatic niche cells (Chen and McKearin, 2003a; Li et al., 2016; Song et al., 2004; Xie and Spradling, 1998, 2000). These ligands activate BMP signaling in GSCs, leading to the transcriptional repression of *bag of marbles* (*bam*), a key gene that promotes differentiation. In contrast, BMP signaling is inactive in cystoblasts, allowing Bam to be expressed and drive their differentiation (Chen and McKearin, 2003a; Song et al., 2004). Bam carries out this function in collaboration with its partner, Benign gonial cell neoplasm (Bgcn) (Li et al., 2009; Ohlstein et al., 2000).

GSCs mutant for *bam* or *bgcn* fail to differentiate and instead hyper-proliferate, forming a well-established *Drosophila* germline tumor model (Lavoie et al., 1999; McKearin and Ohlstein, 1995; McKearin and Spradling, 1990; Niki and Mahowald, 2003). Notably, these germline tumor cells competitively displace wild-type GSCs from the niche (Jin et al., 2008). The resulting displacement creates a microenvironment where wild-type GSCs are surrounded by tumor cells, providing an excellent model system to study stem cell behavior in tumor neighborhoods.

Here, we demonstrate that *bam* or *bgcn* mutant germline tumors inhibit the differentiation of neighboring wild-type GSCs by functionally mimicking the stem cell niche. This mechanism may be conserved in mammals, including humans, during tumorigenesis, where malignant cells could similarly disrupt normal stem cell development.

Results

Germline tumors inhibit the differentiation of neighboring wild-type GSCs

Using the *nos>FLP/FRT* or *hs-FLP/FRT* methods we previously established (Zhang et al., 2023; Zhao et al., 2018), we generated *bam* or *bgcn* mutant germline clones in *Drosophila* ovaries (***Figure 1B***). Surprisingly, we found that many wild-type germ cells located outside the niche retained a GSC-like single-germ-cell (SGC) morphology, even when encapsulated within egg chambers (***Figure 1C, D***, ***Figure 1—figure supplement 1***). Under normal conditions, GSCs that exit the niche differentiate into interconnected germline cysts, where germ cells are linked rather than remaining as individual, isolated cells (Fuller and Spradling, 2007; Xie and Spradling, 2000). To rule out the possibility that the SGC phenotype is an artifact caused by GFP expression, we repeated the experiments using RFP and arm-lacZ as alternative mosaic-analysis markers. Consistent results were observed (***Figure 1E, F***), confirming that the phenotype is not attributable to GFP.

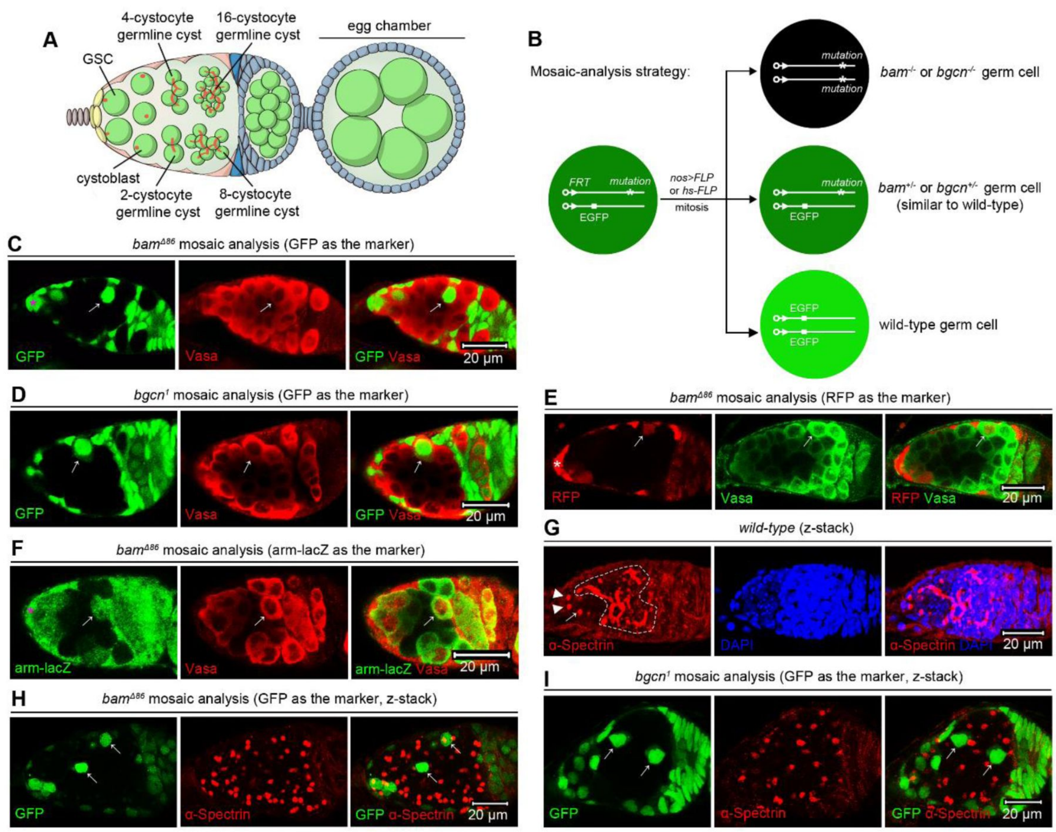

Figure 1.

bam **or** ***bgcn*** **mutant germline tumors inhibit the differentiation of neighboring wild-type GSCs.**

(A) Schematic cartoon for early oogenesis. The red dots and branches indicate spectrosomes and fusomes, respectively. GSC: germline stem cell. (B) Mosaic analysis strategy. The FLP recombinase triggers mitotic recombination by targeting *FRT* sequences. The *nos*>*FLP* method restricts FLP expression to the germline, while the *hs-FLP* method enables heatshock-inducible FLP expression. (C-F) Representative samples. The asterisks mark the niche cap cells, and the arrows indicate SGCs that have exited the niche and are surrounded by germline tumors mutant for *bam* or *bgcn*. Vasa, a germ cell marker, should label all germ cells. However, due to poor tumor permeability, staining often fails to detect tumorous germ cells in the central region (see Vasa panels in D-F). (G-I) Representative samples (z-stack projections). In (G), the arrowheads and arrow respectively mark two GSCs and one cystoblast, all containing dot-like spectrosomes, while the dotted lines delineate cystocytes with branched fusomes. In (H) and (I), the arrows denote SGCs that also contain dot-like spectrosomes, akin to GSCs and the adjacent GSC-like tumor cells. The online version of this article includes the following source data and figure supplement for figure 1: Source data 1. All genotypes. **Figure 1—figure supplement 1** SGCs appear in egg chambers.

To further confirm that these SGCs exhibit GSC-like characteristics, we conducted anti-α-Spectrin immunofluorescent staining, a method that labels a germline-specific organelle known as the spectrosome in GSCs and cystoblasts, and the fusome in cystocytes. GSCs perform complete cell division, whereas cystocytes undergo incomplete cytokinesis, remaining interconnected through fusomes and ring canals. Consequently, spectrosomes appear as dot-like structures, while fusomes exhibit branched morphologies (***Figure 1G***) (Lin et al., 1994). To accurately capture the three-dimensional (3D) architecture of spectrosomes and fusomes, we acquired z-stack images using confocal microscopy. Strikingly, these SGCs displayed dot-like spectrosomes, closely resembling those observed in wild-type GSCs and *bam* or *bgcn* mutant GSC-like tumor cells (***Figure 1H, I***). We also considered the possibility that SGCs might arise through the dedifferentiation of the cystocytes in germline cysts surrounded by germline tumors. If this were the case, such cystocytes would initially undergo complete cell division, leaving behind midbodies as markers of the late cytokinesis stage. When visualized by anti-α-Spectrin immunofluorescence, midbody appears as a central sphere that is slightly connected to two larger flanking structures, resembling a variant of nunchucks (Mathieu et al., 2022). Notably, in our analyses of over 50 germline cysts surrounded by *bam* mutant germline tumors, none contained midbodies, suggesting that dedifferentiation is unlikely to be the primary mechanism responsible for the SGC phenotype. Together, these findings indicate that *bam* or *bgcn* mutant germline tumors inhibit the differentiation of neighboring wild-type GSCs.

The inhibition of differentiation in SGCs relies on the lack of Bam expression

Given that Bam is the key factor promoting GSC differentiation (McKearin and Ohlstein, 1995; Ohlstein and McKearin, 1997), we were very curious about the expression of Bam in SGCs. At first, we assessed Bam protein levels using immunofluorescent staining with an anti-BamC antibody (McKearin and Ohlstein, 1995). Strikingly, none of the SGCs examined (n > 100) were BamC-positive (***Figure 2A***). Then, we analyzed *bam* transcription levels using a *bamP*-GFP reporter (Chen and McKearin, 2003b). Under normal conditions, 100% of GSCs within the niche (n = 153) were GFP-negative, while 98% of cystoblasts (n = 106) were GFP-positive (***Figure 2B, C***), confirming that *bam* transcription is associated with the initiation of GSC differentiation (McKearin and Ohlstein, 1995). Notably, 74% of SGCs (n = 132) were GFP-negative, while the remaining 26% were GFP-positive (***Figure 2B, C***). This suggests that SGCs can be categorized into two distinct groups: those resembling GSCs (GSC-like) and those resembling cystoblasts (cystoblast-like). The cystoblast-like SGCs may have already initiated their differentiation program toward becoming cystocytes. Since *bam* transcription initiates in cystoblasts (McKearin and Spradling, 1990) but Bam proteins accumulate predominantly in cystocytes (McKearin and Ohlstein, 1995), the Bam protein levels in these cystoblast-like SGCs are likely below the detection threshold at this early stage.

Next, we asked whether ectopic expression of Bam can drive SGCs to differentiate. To address this, we established two experimental scenarios: one with the *hs-bam* element and one without as the control. In the *hs-bam* scenario (*with hs-bam*), GFP-positive germ cells are wild-type (carrying *hs-bam*), while GFP-negative cells are *bam*$^{-/-}$ (lacking *hs-bam*). In the control scenario (*without hs-bam*), GFP-positive cells are wild-type, and GFP-negative cells are *bam*$^{-/-}$ (***Figure 2D***, see genotypes in Source data 1). To quantify the SGC phenotype, we calculated the percentage of SGCs relative to the total number of SGCs and germline cysts, considering the out-of-niche germ cells that are either fully enclosed by germline tumors (e.g., the right SGC in ***Figure 1I*** and the marked germline cyst in ***Figure 2G***) or in contact with wild-type germ cells or somatic cells on just one side (e.g., the left SGC in ***Figure 1I*** and the outlined germline cyst in ***Figure 4C***). Although the SGC phenotype was detectable after fly eclosion, we quantified it in 14-day-old flies to ensure robust germline tumor growth, improving quantification efficiency. To induce ectopic Bam expression, 12-day-old female flies were subjected to heatshock treatment, which involved heating at 37°C for two hours, twice daily with a 6-hour interval, and over two consecutive days. In

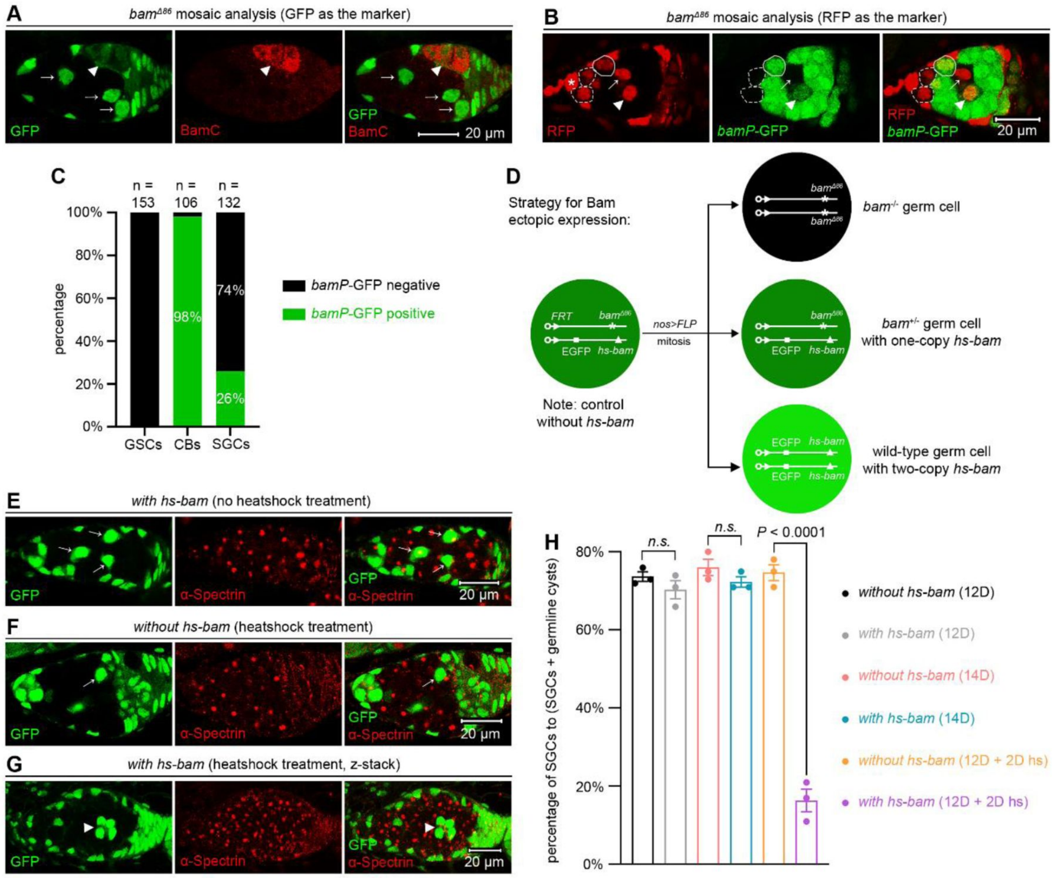

Figure 2.

The inhibition of SGC differentiation depends on the lack of Bam expression.

(A) Representative sample. The arrowhead marks a BamC-positive 4-cystocyte germline cyst, while the arrows indicate BamC-negative SGCs. (B) Representative sample. The asterisk denotes niche cap cells, and the dotted circles outline *bamP*-GFP-negative GSCs. The solid circle marks a *bamP*-GFP-positive cystoblast. The arrow and arrowhead point to *bamP*-GFP-negative and -positive SGCs, respectively. (C) Quantification data. 14-day-old flies were used for the analyses. CBs: cystoblasts. (D) Schematic of the experimental strategy for (E-H). In "*with hs-bam*" flies (E and G), wild-type germ cells (both *bam*$^{+/+}$ and *bam*$^{+/-}$) carry the *hs-bam* transgene, while control "*without hs-bam*" flies (F) lack this element in their wild-type germ cells. (E-G) Representative samples. The arrows mark SGCs with dot-like spectrosomes, while the arrowhead indicates a 4-cystocyte germline cyst containing branched fusomes. (H) Quantification data. For each experiment, three independent replicates were performed, with over 100 SGCs and germline cysts quantified per replicate. Data represent mean ± SEM, and statistical significance was determined by t test. *n.s.* ($P > 0.05$). The online version of this article includes the following source data for **figure 2**: **Source data 1.** All genotypes. **Source data 2.** Raw quantification data.

the absence of heatshock treatment, the percentage of SGCs in ovaries of both genotypes showed no significant difference at either 12 or 14 days (***Figure 2E-H***), indicating that the *hs-bam* element alone, without heatshock, does not affect the phenotype. However, following heatshock treatment, the percentage of SGCs in ovaries with *hs-bam* was markedly reduced compared to those without *hs-bam* (***Figure 2E-H***), suggesting that ectopic Bam expression can drive SGCs to differentiate. Collectively, these results support that the differentiation defects of SGCs are due to the lack of Bam expression.

SGCs retain moderate BMP signaling activation

Within the niche, BMP signaling functions to repress *bam* transcription to inhibit GSC differentiation (Chen and McKearin, 2003a; Song et al., 2004). To investigate BMP signaling activation in SGCs, we employed immunofluorescent staining for pMad, a well-characterized marker of BMP signaling activity (Kai and Spradling, 2003). Surprisingly, we observed undetectable pMad levels in all SGCs examined (n > 100) (***Figure 3A, B***). To investigate this further, we examined the activity of *Dad*-lacZ, a highly sensitive BMP signaling reporter known to be activated not only in GSCs but also in cystoblasts (Kai and Spradling, 2003; Song et al., 2004). Notably, 73% of SGCs were lacZ-positive (n = 107), a proportion lower than that of GSCs within the niche, which showed 100% lacZ positivity (n = 122) (***Figure 3C, D***). Furthermore, when comparing *Dad*-lacZ expression levels exclusively in lacZ-positive cells, we found that SGCs exhibited significantly lower expression levels than GSCs within the niche (***Figure 3C, E***). These findings indicate that BMP signaling is activated in SGCs but at lower levels than those in GSCs within the niche.

Beyond maintaining *Drosophila* female GSCs in the niche, BMP signaling also promotes their division (Xie and Spradling, 1998). Since the activation levels of BMP signaling in SGCs were lower than those in GSCs within the niche, we hypothesized that SGCs would exhibit slower proliferation rates than GSCs. To test this hypothesis, we performed BrdU incorporation assays. The results revealed that only 4.5% of SGCs were BrdU-positive (n = 1034), a significantly lower proportion than the 7.8% observed in GSCs within the niche (n = 1337) (***Figure 3F-H***). These findings further corroborate the reduced activation of BMP signaling in SGCs relative to GSCs.

BMP signaling inhibits SGC differentiation

Then we investigated whether BMP signaling functions to inhibit SGC differentiation. The BMP type II receptor Punt and the co-Smad Medea (Med) are essential for maintaining GSC stemness within the niche (Xie and Spradling, 1998). To investigate whether they are also required to inhibit SGC differentiation, we established a genetic scenario, in which GFP$^{+/+}$ RFP$^{-/-}$ germ cells are *punt*$^{-/-}$or *med*$^{-/-}$; GFP$^{+/-}$ RFP$^{+/-}$ germ cells are *punt*$^{+/-}$ *bam*$^{+/-}$ or *med*$^{+/-}$ *bam*$^{+/-}$(similar to wild-type); and GFP$^{-/-}$ RFP$^{+/+}$ germ cells are *bam*$^{-/-}$. In control experiments (with no *punt* or *med* mutation), GFP$^{+/+}$ RFP$^{-/-}$ germ cells are wild-type; GFP$^{+/-}$ RFP$^{+/-}$ germ cells are *bam*$^{+/-}$ (similar to wild-type); and GFP$^{-/-}$ RFP$^{+/+}$ germ cells are *bam*$^{-/-}$(***Figure 4A***, see genotypes in Source data 1). Strikingly, the proportion of *punt*$^{-/-}$ or *med*$^{-/-}$SGCs relative to total SGCs was significantly lower than in controls (***Figure 4B-E***). Conversely, among *punt*$^{-/-}$ or *med*$^{-/-}$germ cells meeting our established criteria for SGC phenotype quantification, germline cysts constituted a higher percentage compared to controls (***Figure 4F***). These results indicate that Punt and Med function to inhibit SGC differentiation.

Mothers against dpp (Mad) is the primary transcription factor of BMP signaling, and it is also essential for GSC maintenance in *Drosophila* ovaries (Xie and Spradling, 1998). Unlike *punt* and *med*, which reside on the same chromosome arm (3R) as *bam*, *mad* is located on a separate chromosome arm (2L). To investigate whether Mad is required to inhibit SGC differentiation, we established a genetic scenario, in which GFP$^{-/-}$ germ cells are *mad*$^{-/-}$ and GFP$^{+/+}$ germ cells are *bam*$^{-/-}$. In control experiments (with no *mad* mutation), GFP$^{-/-}$ germ cells are wild-type and GFP$^{+/+}$ germ cells are *bam*$^{-/-}$ (***Figure 4G***, see genotypes in Source data 1). Notably, *mad* mutation

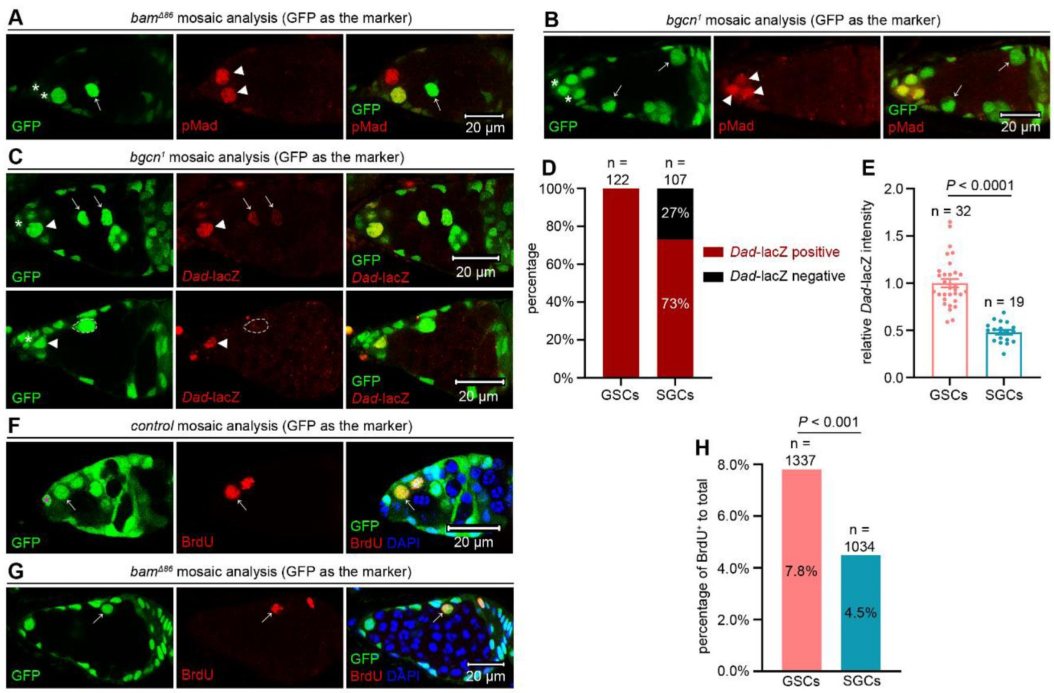

Figure 3.

SGCs maintain lower BMP signaling levels than GSCs within the niche.

(A and B) Representative samples. The asterisks mark niche cap cells, arrowheads indicate pMad-positive GSCs, and arrows point to pMad-negative SGCs. (C) Representative samples. The asterisks denote niche cap cells, arrowheads mark *Dad*-lacZ-positive GSCs, and arrows highlight *Dad*-lacZ-positive SGCs. The dotted cycles outline one *Dad*-lacZ-negative SGC. (D and E) Quantification data. 14-day-old flies were used for the analyses. In (E), data represent mean ± SEM, and statistical significance was determined by t test. (F) Representative sample. The asterisk marks the niche cap cell, while the arrows indicate a BrdU^{+} GSC within the niche. (G) Representative sample. The arrow indicates a BrdU^{+} SGC surrounded by germline tumors. (H) Quantification data. 14-day-old flies were used for the analyses. Statistical significance was determined by chi-squared test.

The online version of this article includes the following source data for **figure 3**: **Source data 1.** All genotypes. **Source data 2.** Raw quantification data.

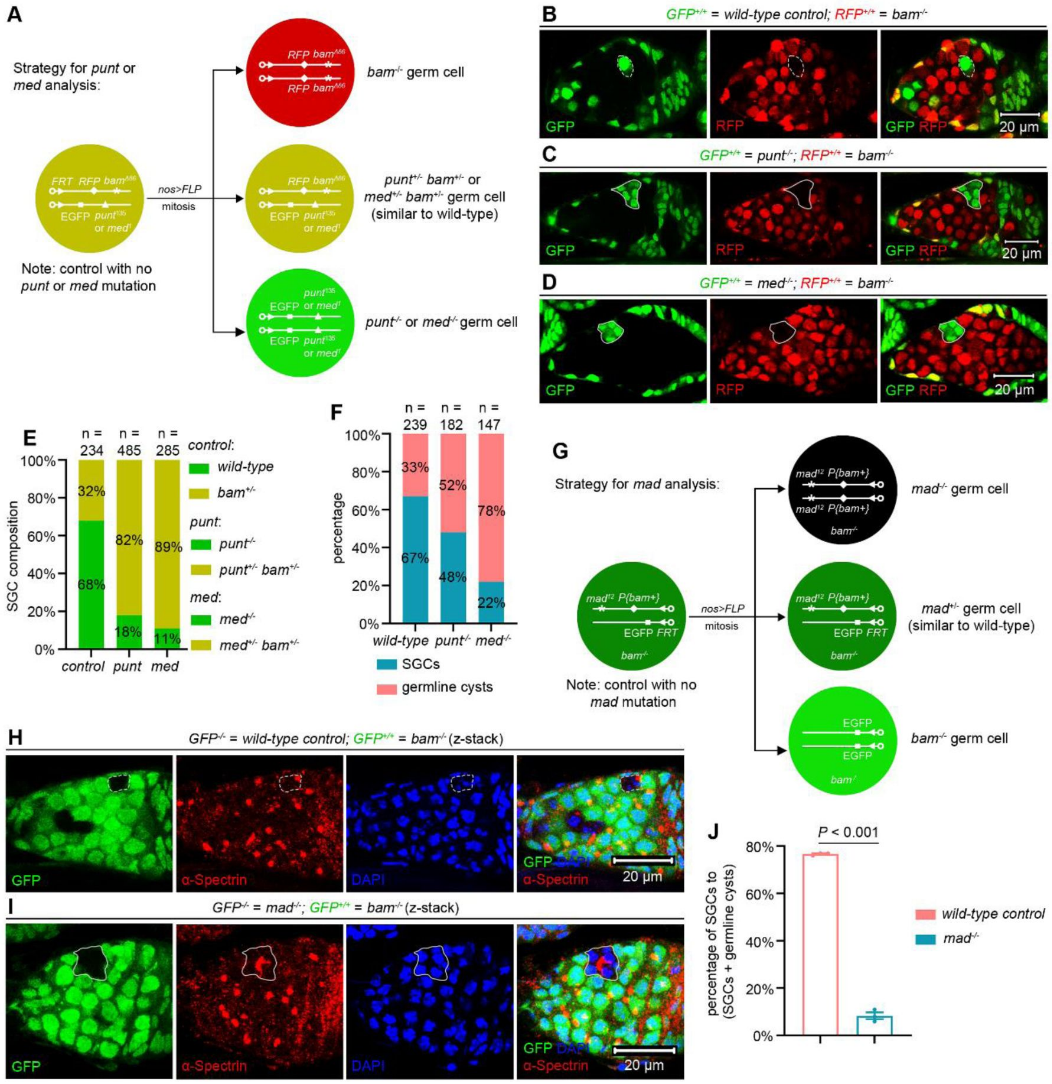

Figure 4.

BMP signaling inhibits SGC differentiation.

(A) Schematic of the experimental strategy for (B-F). Genotypes were unambiguously distinguished using a triple-color system (red, yellow, and green). (B-D) Representative samples. The dotted cycles mark an SGC, while the solid lines outline germline cysts containing differentiating cystocytes. (E and F) Quantification data. 14-day-old flies were used for the analyses. (G) Schematic of the experimental strategy for (H-J). (H and I) Representative samples. The dotted lines mark an SGC, while the solid lines outline a germline cyst containing differentiating cystocytes. (J) Quantification data. 14-day-old flies were used for the analyses. For each experiment, three independent replicates were performed, with over 100 SGCs and germline cysts quantified per replicate. Data represent mean ± SEM, and statistical significance was determined by t test. The online version of this article includes the following source data for **figure 4**: **Source data 1.** All genotypes. **Source data 2.** Raw quantification data.

significantly decreased the SGC proportion relative to controls (***Figure 4H-J***). These results suggest that, like Punt and Med, Mad also plays a crucial role in suppressing SGC differentiation. Together, these findings demonstrate that BMP signaling contributes to inhibiting SGC differentiation, despite at reduced activation levels.

Germline tumors secret Dpp and Gbb

The next question we sought to address was which cells surrounding SGCs secrete BMP ligands, such as Dpp and Gbb. Guided by the Occam's Razor principle, we focused on *bam* or *bgcn* mutant germline tumor cells, as the SGC phenotype is dependent on their presence. Initial attempts to detect expression of these two ligand genes using *in situ* hybridization proved unsuccessful. We therefore employed an established *dpp*-lacZ (*P4*-lacZ) reporter (Li et al., 2016) to monitor Dpp expression. Consistent with previous findings, we observed robust lacZ signals in cap cells (***Figure 5A***), the primary niche cells that are responsible for BMP ligand secretion (Chen and McKearin, 2003a; Song et al., 2004; Xie and Spradling, 2000). Notably, *bam* mutant germline tumor cells displayed lower lacZ expression levels than cap cells but significantly higher levels than differentiating cystocytes (***Figure 5A-C***).

To more sensitively assess Dpp and Gbb expression, we performed real-time quantitative PCR (RT-qPCR) analyses in *bam* or *bgcn* mutant ovaries, comparing samples with and without germline-specific knockdown of *dpp* or *gbb*. Detection of reduced transcript levels in knockdown conditions would confirm active expression of these genes in the respective genetic backgrounds. Consistent with the essential roles of these two genes in fly viability, ubiquitous knockdown using *act-GAL4* with either *dpp-RNAi* or *gbb-RNAi* caused lethality, which results also validated the efficacy of these RNAi lines. Notably, germline-specific knockdown of *dpp* or *gbb* significantly reduced their transcript levels compared to *yellow* (*y*) or *white* (*w*) knockdown controls (***Figure 5D, E***). These findings demonstrate that *bam* or *bgcn* mutant germline tumors secrete the BMP ligands, albeit at lower levels than cap cells.

Dpp and Gbb secreted by germline tumors are required to inhibit SGC differentiation

Finally, we investigated whether Dpp and Gbb secreted by germline tumors are required to inhibit SGC differentiation. Using a previously established double-mutant mosaic-analysis strategy for two genes on different chromosomes (Zhang et al., 2024; Zhang et al., 2023), we generated *dpp bam* or *gbb bam* double-mutant germline clones using two *dpp* mutant alleles, dpp^{d6}, dpp^{d12}, and one *gbb* allele, gbb^{1} (***Figure 6A, B***, see genotypes in Source data 1). Heterozygotes in any of these alleles did not affect GSC maintenance, germ cell differentiation, and female fly fertility (***Figure 6—figure supplement 1***). However, both *dpp bam* and *gbb bam* double-mutant germline tumor cells exhibited reduced proliferation rates compared to *bam* single-mutant controls (***Figure 6—figure supplement 2***), indicating that autocrine BMP signaling promotes *bam* mutant tumor growth. As mentioned earlier, our evaluation focused on germ cells that have exited the niche and are surrounded by germline tumors to quantify the SGC phenotype. However, this raises the question of whether the extent of tumor encirclement (i.e., being surrounded by more or fewer tumor cells) influences the phenotype. To investigate this, we compared the SGC phenotype in bigger and smaller *bam* mutant germline tumors. Using confocal microscopy, we analyzed 70 germaria containing *bam* mutant germline clones that met our criteria for SGC phenotype quantification. For each *bam* mutant germline clone, we scanned and measured its maximal 2D cross-sectional area as a proxy for its 3D size. The 35 bigger and 35 smaller clones were categorized as 'bigger' and 'smaller' tumors, respectively. Strikingly, we observed no significant difference in the SGC phenotype between these two groups of tumors (***Figure 6—figure supplement 3***), suggesting that direct contact between tumorous and wild-type germ cells, rather than tumor size, is the primary determinant of this phenotype.

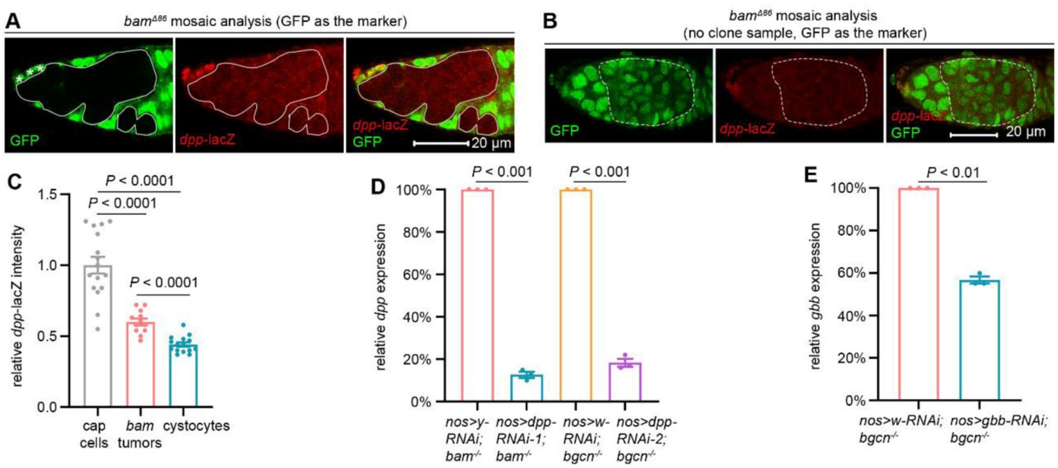

Figure 5.

Germline tumors secret the BMP ligands.

(A and B) Representative samples. In (A), the asterisks denote *dpp*-lacZ-high niche cap cells, and solid lines outline GFP-negative *bam* mutant germline tumor cells. In (B), the dotted lines highlight GFP-positive wild-type cystocytes. (C) Quantification data. 14-day-old flies were used for the analyses. (D and E) Quantification data for RT-qPCR assays. 14-day-old flies were used for the analyses. For each experiment, three independent replicates were performed. Statistical significance in (C) was determined by one-way ANOVA and in (D and E) by t test. The online version of this article includes the following source data for **figure 5**: **Source data 1.** All genotypes. **Source data 2.** Raw quantification data.

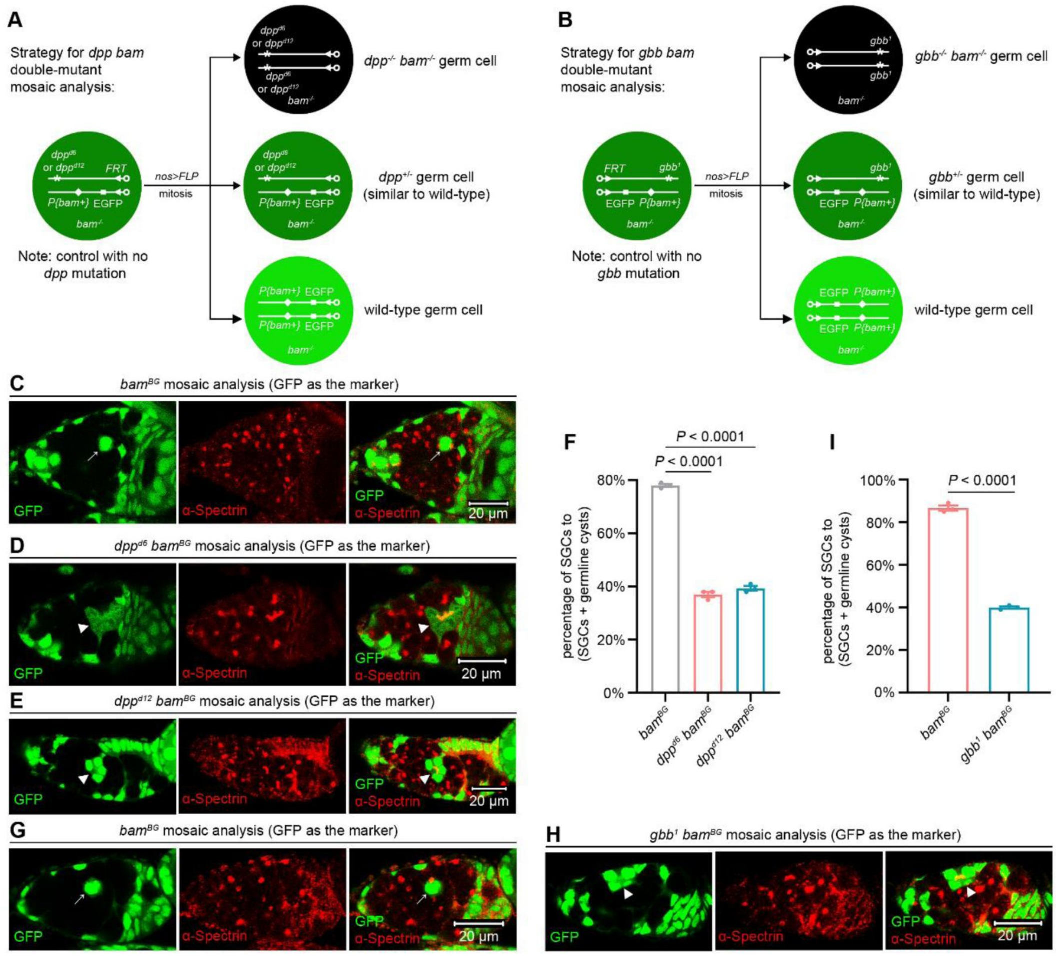

Figure 6.

Dpp and Gbb secreted by germline tumors are required to inhibit SGC differentiation.

(A) Schematic of the experimental strategy for (C-F). (B) Schematic of the experimental strategy for (G-I). (C-E, G, and H) Representative samples. The arrows mark SGCs containing dot-like spectrosomes, while the arrowheads denote germline cysts with differentiating cystocytes that possess branched fusomes. (F and I) Quantification data for the SGC phenotype. 14-day-old flies were used for the analyses. For each experiment, three independent replicates were performed, with over 100 SGCs and germline cysts quantified per replicate. Data represent mean ± SEM. Statistical significance in (F) was determined by one-way ANOVA and in (I) by t test. The online version of this article includes the following source data and figure supplement for figure 6: **Source data 1.** All genotypes. **Source data 2.** Raw quantification data. **Figure 6—figure supplement 1** Monoallelic deletion of *dpp* or *gbb* does not affect GSC maintenance, germ cell differentiation, and female fly fertility. **Figure 6—figure supplement 2** *dpp bam* or *gbb bam* double-mutant germline tumor cells divide more slowly than *bam* single-mutant ones. **Figure 6—figure supplement 3** The SGC phenotype is unchanged irrespective of the number of surrounding germline tumors.

The results above demonstrate that comparing the severity of the SGC phenotype is feasible between germ cells surrounded by smaller *dpp bam* or *gbb bam* double-mutant germline tumors and those surrounded by larger *bam* single-mutant germline tumors. Remarkably, both *dpp bam* and *gbb bam* double-mutant germline tumors enclosed fewer SGCs but more germline cysts than their *bam* single-mutant counterparts (***Figure 6C-I***). Thus, we concluded that the BMP ligands from directly-contacting germline tumor cells mediate the dominant inhibition of SGC differentiation. This amazingly parallels the mechanism observed in normal stem cell niche, where only germ cells in direct contact with cap cells are maintained as GSCs (Chen and McKearin, 2003a; Song et al., 2004; Xie and Spradling, 2000).

Discussion

Our study reveals that *bam* or *bgcn* mutant germline tumors in *Drosophila* ovaries secrete lower levels of BMP ligands Dpp and Gbb than cap cells, resulting in moderate BMP signaling activation in adjacent wild-type GSCs (called SGCs in this study). Such BMP signaling activation is sufficient to repress *bam* transcription, thereby blocking SGC differentiation (see our working model in ***Figure 7***). Strikingly, this mechanism closely recapitulates the normal niche signaling program mediated by cap cells (Chen and McKearin, 2003a; Song et al., 2004; Xie and Spradling, 1998, 2000). To our knowledge, this represents the first evidence that tumor cells can functionally mimic a stem cell niche to arrest neighboring wild-type stem cells in an undifferentiated state.

bam or *bgcn* mutant germline tumors consist of GSC-like cells (Lavoie et al., 1999; McKearin and Ohlstein, 1995), which are expected to resemble SGCs. However, out-of-niche *bgcn* mutant germline tumor cells did not activate BMP signaling to the same extent as SGCs, as evidenced by much lower *Dad*-lacZ expression in GFP-negative *bgcn* mutant tumor cells than in neighboring GFP-positive SGCs (***Figure 3C***). Consistent with this, while only 26% of SGCs expressed *bamP*-GFP, most of out-of-niche *bam* mutant germline tumor cells were *bamP*-GFP-positive (***Figure 2B, C***), further supporting reduced BMP signaling in these tumor cells relative to SGCs. These findings suggest that SGCs are more responsive to BMP signals secreted by germline tumors than the tumors themselves. Future studies are needed to elucidate the underlying mechanisms.

One interesting finding is that *bam* or *bgcn* mutant germline tumors secrete lower levels of BMP ligands than cap cells (***Figure 5A-C***). This aligns with earlier microarray data showing that purified *Drosophila* female GSCs express minimal Dpp and Gbb (Kai et al., 2005). However, our work reveals that such BMP levels in germline tumors are functionally critical to suppress SGC differentiation (***Figure 6***). Unlike normal GSCs, which receive unidirectional BMP ligands from cap cells (Chen and McKearin, 2003a; Li et al., 2016; Song et al., 2004; Xie and Spradling, 2000), SGCs are often fully surrounded by *bam* or *bgcn* mutant germline tumors. This spatial advantage likely enables tumors to inhibit SGC differentiation efficiently without matching the high BMP output of cap cells. Moreover, since BMP signaling is known to both inhibit normal GSC differentiation and promote their proliferation (Xie and Spradling, 1998), it should similarly stimulate SGC expansion, which is detrimental for *bam* or *bgcn* mutant germline tumors. We propose that these tumor cells finely regulate BMP secretion to balance these opposing demands: maintaining differentiation blockade of SGCs while avoiding stimulation of their excessive proliferation.

A well-established principle in oncology is that tumor aggressiveness correlates with poor differentiation, with less-differentiated tumors exhibiting enhanced transformative capacity and metastatic potential (Jogi et al., 2012; Lytle et al., 2018). In *Drosophila* ovaries, *bam* or *bgcn* mutant germline tumors consist of GSC-like cells that may resemble these poorly differentiated human tumors (Lavoie et al., 1999; McKearin and Ohlstein, 1995). This similarity raises the possibility that stem cell-like human tumors may similarly inhibit the differentiation of adjacent wild-type stem cells. By blocking differentiation, such tumors could deplete terminally

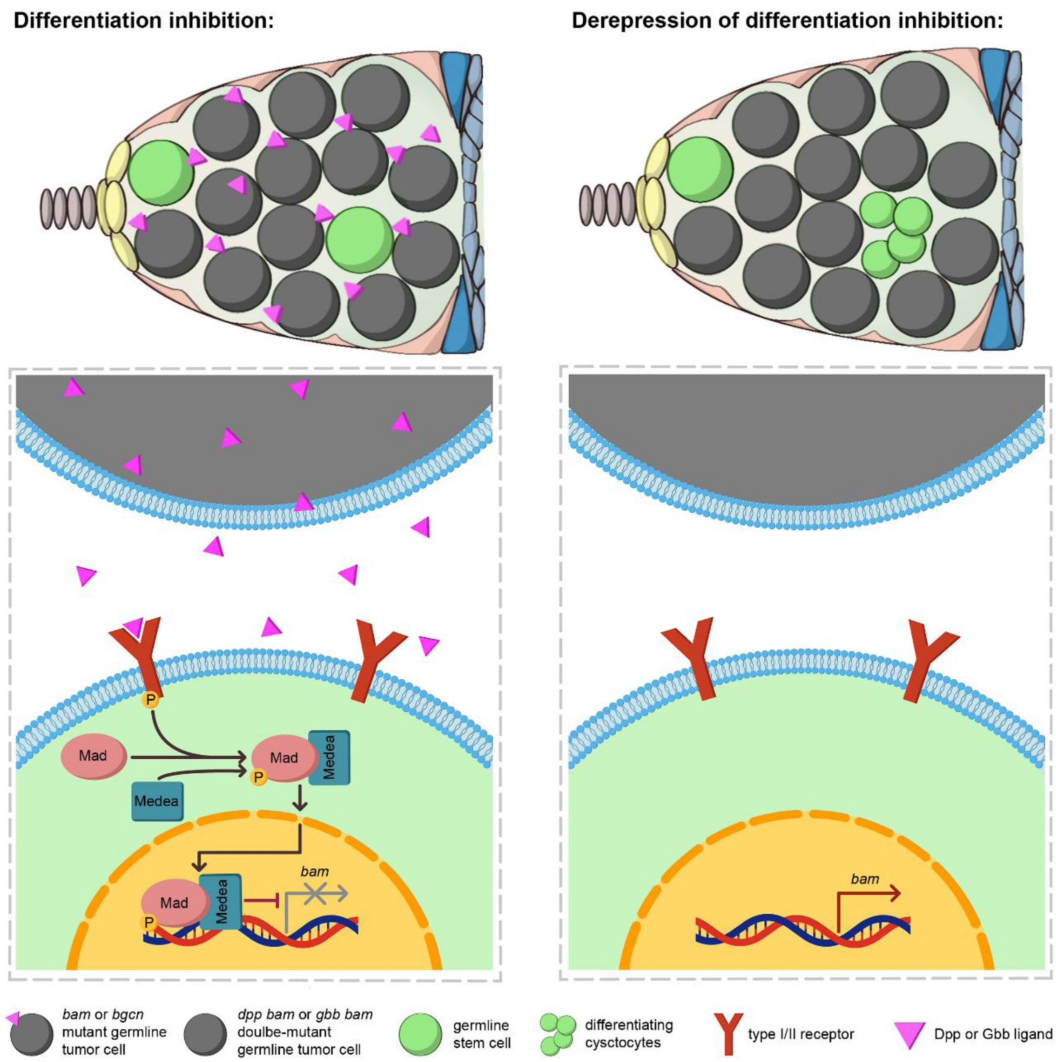

Figure 7.

A working model.

bam or *bgcn* mutant germline tumors secrete the BMP ligands Dpp and Gbb to activate BMP signaling in out-of-niche GSCs (called SGCs in this study) to inhibit their differentiation (left panel). In contrast, *dpp bam* and *gbb bam* double-mutant germline tumors exhibit a significant loss of this differentiation-inhibiting ability (right panel).

differentiated cell populations, potentially exacerbating patient mortality. This mechanism may contribute to the heightened lethality of poorly differentiated tumors. Further investigation is needed to test this hypothesis.

The differentiation of a single GSC into a 16-cell germline cyst, comprising 15 polyploid nurse cells and one developing oocyte, represents a substantial metabolic investment (Fuller and Spradling, 2007; Lin, 1997). We propose that *bam* or *bgcn* mutant germline tumors block this process to divert nutrients toward their own uncontrolled growth. This phenomenon could have broad implications, as many human tissues and organs (intestine, muscle, skin, blood system, male germline, etc.) similarly depend on adult stem cells for homeostasis (Blanpain and Fuchs, 2006; Gehart and Clevers, 2019; Sousa-Victor et al., 2022; Spradling et al., 2011; Wilkinson et al., 2020). Notably, these stem cell-dependent tissues and organs are frequent sites of tumorigenesis, raising the possibility that human cancers may similarly impair neighboring stem cell differentiation to optimize nutrient allocation for malignant growth. A key limitation of our study is that the evidence is derived solely from *Drosophila* germline. Future work should explore whether similar regulatory paradigms operate in mammalian tissues during tumorigenesis.

Materials and Methods

Key resources table

Reagent or Resource	Source	Identifier
Antibodies		
mouse anti-α-Spectrin	Developmental Studies Hybridoma Bank, DSHB	3A9
mouse anti-BamC	DSHB	AB_10570327
mouse anti-β-Gal	DSHB	JIE7
mouse anti-BrdU	Sigma	B5002
rabbit anti-pMad	Ed Laufer	N/A
rabbit anti-Vasa	Zhaohui Wang	(***Chen et al., 2014***)
goat anti-mouse 546	Invitrogen	A11003
goat anti-rabbit 488	Apexbio	K1206
goat anti-rabbit 546	Invitrogen	A11035
cDNA clone		
bam cDNA clone	Berkeley *Drosophila* Genome Project, BDGP	FI05606
***Drosophila* strains**		
act-GAL4	Zhaohui Wang	N/A
bam^{BG} (strong loss-of-function allele)	Zhaohui Wang	(***Chen and McKearin, 2005***)
$bam^{\Delta 86}$ (null allele)	Michael Buszczak	(***Bopp et al., 1993***)

bamP-GFP	Ting Xie	(***Chen and McKearin, 2003b***)
bgcn^{1} (null allele)	Bloomington *Drosophila* Stock Center, BDSC	6054
Canton-S	BDSC	64349
Dad-lacZ	Zheng Guo	(***Kai and Spradling, 2003***)
dpp^{d6} (hypomorphic allele)	BDSC	2062
dpp^{d12} (hypomorphic allele)	BDSC	2070
dpp(P4)-lacZ	Rongwen Xi	(***Li et al., 2016***)
EGFP FRT40A	BDSC	5629
FRT40A	BDSC	1816
FRT42D	BDSC	1802
FRT42D EGFP	BDSC	5626
FRT82B	BDSC	86313
FRT82B arm-lacZ	BDSC	7369
FRT82B EGFP	BDSC	32655
FRT82B RFP	BDSC	30555
gbb^{1} (null allele)	BDSC	98344
hs-bam on chromosome 3R	this study	N/A
mad^{12} (null allele)	BDSC	51301
med^{1} (null allele)	BDSC	9033
nos-GAL4-VP16	Ruth Lehmann	(***Van Doren et al., 1998***)
P{bam+}	Shaowei Zhao	(***Zhang et al., 2023***)
punt^{135} (strong loss-of-function allele)	BDSC	3100

UASp-dpp-RNAi-1	TsingHua Fly Center, THFC	TH201500984.S
UASp-dpp-RNAi-2	THFC	THU5880
UASp-FLP	Shaowei Zhao	(***Zhang et al., 2023***)
UASp-gbb-RNAi	THFC	THU1480
UASp-GFP	Michael Buszczak	N/A
UASp-yellow-RNAi	THFC	TH03150.N
UASz-FLP	Shaowei Zhao	(***Zhang et al., 2023***)
Kits		
ChamQ SYBR qPCR Master Mix	Vazyme	Q311
HiFiScript cDNA Synthesis Kit	CWBIO	CW2569M
RNeasy Micro Kit	Qiagen	74004
Plasmids		
bam cDNA clone	Berkeley *Drosophila* Genome Project	FI05606
pCaSpeR-hs	Eric Baehrecke	N/A
attB-pCaSpeR-hs-bam	this study	N/A
Software		
Adobe Photoshop 2022	San Jose, CA, USA	
ImageJ	NIH	
Prism	Graphpad Software, Inc.	
ZEN 3.0 SR imaging software	Carl Zeiss	

Data availability

All genotypes are described in Source data 1, and the raw quantification data are included in Source data 2.

Fly husbandry

Flies were raised at 25°C on standard cornmeal/molasses/agar media.

Transgenic flies

hs-bam on chromosome 3R

The coding sequence of the *bam* gene, amplified from the cDNA clone, was cloned into the *BglII-XbaI* sites of the *pCaSpeR-hs* vector (a gift from Eric Baehrecke), while the *attB* sequence was inserted into the *XhoI* site. The resulting *attB-pCaSpeR-hs-bam* plasmid was then microinjected into the *attP154* (Chromosome 3R, *97D2*) fly strain to generate site-specific transgenic flies.

Heatshock method to induce germline clones

To ensure developmental synchrony and maintain low-density growth, eggs within 8 hours of laying were collected for heatshock treatment. The animals (late-Larva 3/early-Pupa stage) were subjected to twice-daily heatshocks at 37°C (2 hours per session, with a 6-hour interval between the two sessions) for 6 consecutive days.

Fertility test

For each genotype, three independent crosses were performed. Each cross vial contained two females and four w^{1118} (wild-type) males, all aged three days old. The crosses were transferred to fresh vials every two days, with five replicate vials quantified per genotype. After all adult flies eclosed, offspring production was assessed by counting the number of empty pupae on the vial walls.

BrdU labeling

Ovaries were dissected in Schneider's insect medium (SIM) and incubated in freshly prepared BrdU solution (100 µg/mL in SIM) for five hours at 25°C. After washing with PBS (30 min), samples were fixed in 4% paraformaldehyde (PBS-diluted) for three hours, followed by another PBS wash (30 min). Samples were then treated with RQ1 DNase reaction solution (Promega, Madison, WI, USA) for one hour, washed with PBST (0.3% Triton X-100 in PBS, 30 min), and incubated overnight at 4°C with mouse anti-BrdU antibody. Following a PBST wash (one hour), ovaries were incubated with goat anti-mouse 546 and DAPI (0.1 µg/mL) in PBST for three hours, washed again in PBST (one hour), and mounted in 70% glycerol (autoclaved).

Immuno-florescent staining, image collection, and data Processing

Ovaries were dissected in PBS, fixed in 4% paraformaldehyde (in PBS) for three hours, washed with PBST (0.3% Triton X-100 in PBS) for 30 min, and then incubated overnight at 4°C with primary antibodies diluted in PBST as follows: mouse anti-BamC (1:5), mouse anti-β-Gal (1:200), mouse anti-α-Spectrin (1:100), rabbit anti-pMad (1:500), and rabbit anti-Vasa (1:2000). After washing with PBST (one hour), samples were incubated with Alexa Fluor-conjugated secondary antibodies (1:1000) and 0.1 µg/mL DAPI (in PBST) for three hours, followed by a final PBST wash (one hour). Ovaries were mounted in 70% glycerol and imaged using a Zeiss LSM 710 confocal microscope (Carl Zeiss AG, Baden-Württemberg, Germany). Images were processed with ZEN 3.0 SR imaging software (Carl Zeiss) and Adobe Photoshop 2022. The quantification data were processed by Microsoft Excel or GraphPad Prism.

RT-qPCR

Ovaries from 14-day-old flies were dissected, and total RNA was extracted using the RNeasy Micro Kit. Equal amounts of RNA were reverse-transcribed into cDNA using the HiFiScript cDNA Synthesis Kit. RT-qPCR was performed on a CFX Connect Real-Time PCR System (Bio-Rad) with ChamQ SYBR qPCR Master Mix. The PCR protocol consisted of an initial denaturation at 95°C for 30 min, followed by 40 cycles of 95°C for 10 sec and 60°C for 30 sec. Relative gene expression was calculated using the $2^{-\Delta\Delta CT}$ method (Livak and Schmittgen, 2001). The primers described previously were used (Huang et al., 2017): *dpp* primer-1: 5'-TACCACGCCATCCACTCAAC-3' *dpp* primer-2: 5’-GCTCGTTACTCGATACGGCT-3’ *gbb* primer-1: 5’-CTGGATCATCGCACCAGAGG-3’ *gbb* primer-2: 5’-GTCTGGACGATCGCATGGTT-3’, *rp49* (internal control) primer-1: 5’-CACCGGATTCAAGAAGTTCC-3’ *rp49* (internal control) primer-2: 5’-GACAATCTCCTTGCGCTTCT-3’

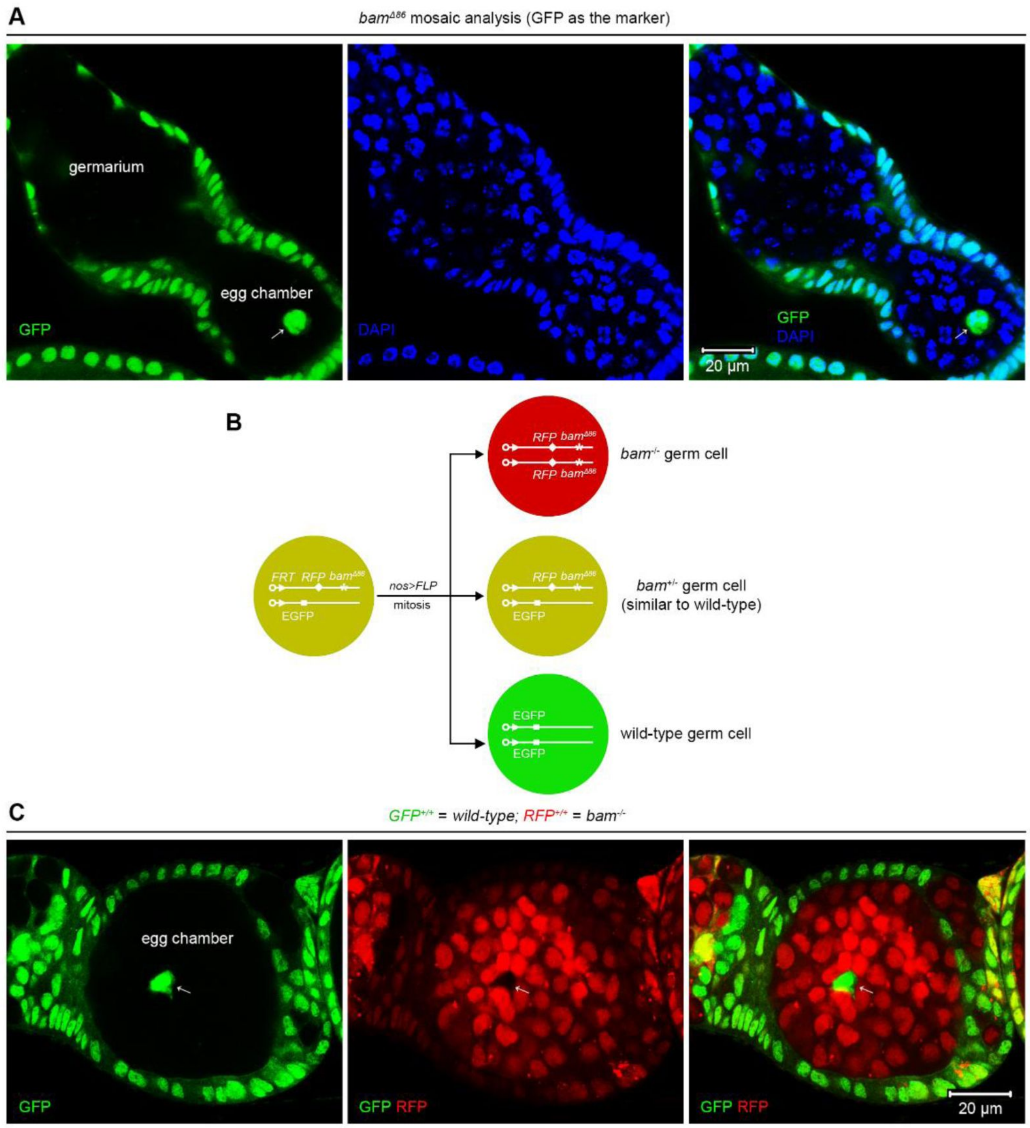

Figure 1-figure supplement 1.

SGCs appear in egg chambers.

(A, C) Representative samples. The arrows indicate SGCs enclosed within egg chambers. (B) Schematic of the experimental strategy for (C). See also Source data 1.

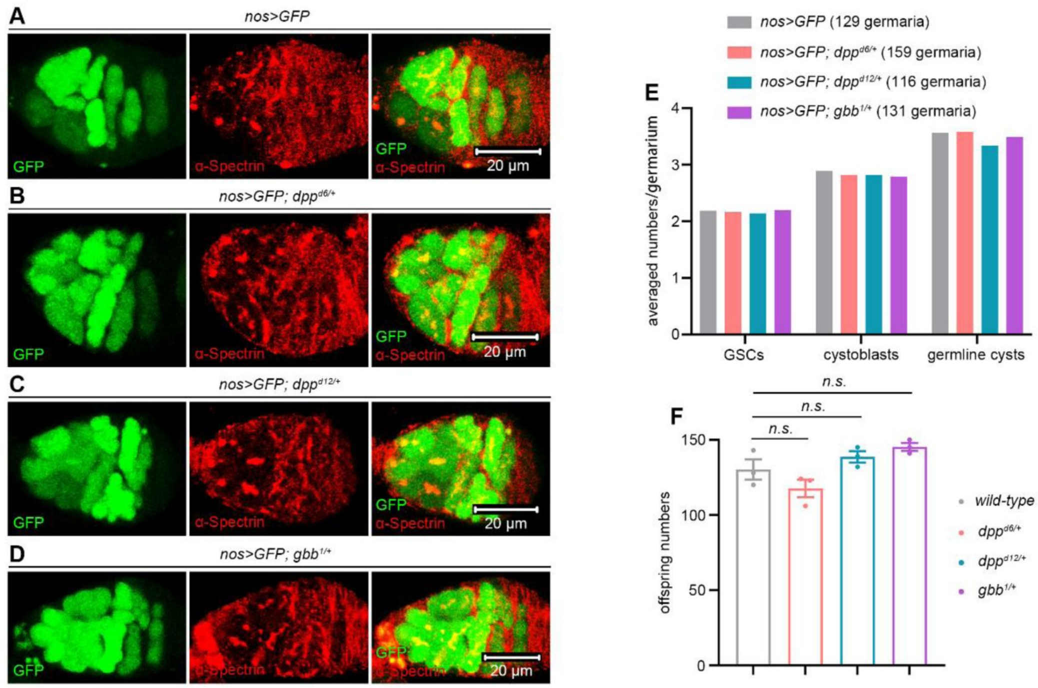

Figure 6-figure supplement 1.

Monoallelic deletion of *dpp* or *gbb* does not affect GSC maintenance, germ cell differentiation, and female fly fertility.

(A-D) Representative samples. (E) Quantification data. 14-day-old flies were used for the analyses. GSCs are germ cells that are located within the niche and contain dot-like spectrosomes. Cystoblasts are germ cells that have exited the niche, remain in contact with GSCs, and maintain dot-like spectrosomes. Cystocytes in germline cysts are germ cells that are characterized by branched fusomes. (F) Fertility test. 3-day-old flies were used for the analyses. For each genotype, three independent replicates were performed. Data represent mean ± SEM, and statistical significance was determined by one-way ANOVA. *n.s.* ($P > 0.05$). See also Source data 1 and 2.

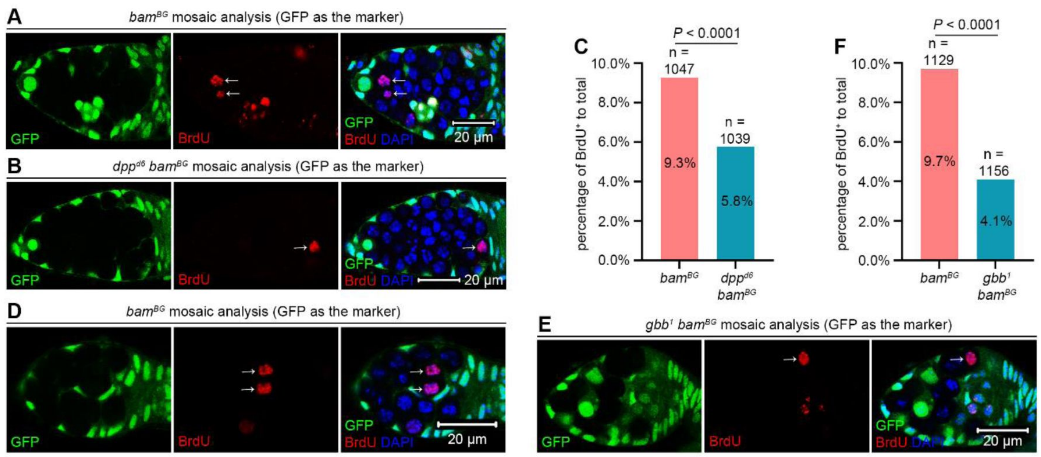

Figure 6-figure supplement 2.

dpp bam or gbb bam double-mutant germline tumor cells divide more slowly than bam single-mutant ones.

(A, B, D, and E) Representative samples. The arrows indicate $BrdU^+$ germline tumor cells mutant for *bam*, *dpp bam*, or *gbb bam*. (C and F) Quantification data. 14-day-old flies were used for the analyses. Statistical significance was determined by chi-squared test. See also Source data 1 and 2.

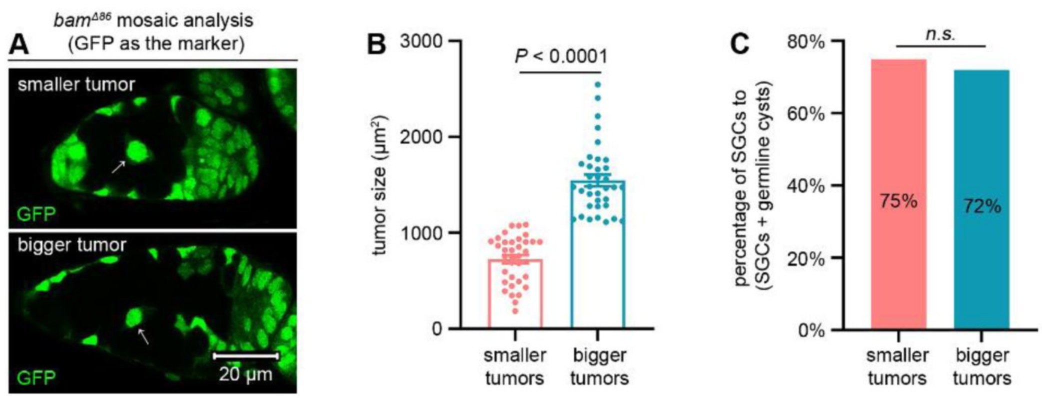

Figure 6-figure supplement 3.

The SGC phenotype is unchanged irrespective of the number of surrounding germline tumors.

(A) Representative samples. The arrows denote SGCs. Both images are of the same magnification. (B) Quantification data for tumor size. Data represent mean ± SEM. (C) Quantification data for the SGC phenotype. In (B and C), 14-day-old flies were used for the analyses. Statistical significance in (B) was determined by t test and in (C) by chi-squared test. *n.s.* ($P > 0.05$). See also Source data 1 and 2.

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Additional information

Author contributions

S.Z. conceived and supervised this study. Y.Z., Y.W., J.S., L.Y., Z.W., D.S., Y.D.Z., and S.Z. performed the experiments. S.Z. wrote the manuscript and all authors commented on it.

Additional files

Source data 1. All genotypes.

Source data 2. Raw quantification data.

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Reviewer #1 (Public review):

Summary:

This preprint from Shaowei Zhao and colleagues presents results that suggest tumorous germline stem cells (GSCs) in the Drosophila ovary mimic the ovarian stem cell niche and inhibit the differentiation of neighboring non-mutant GSC-like cells. The authors use FRT-mediated clonal analysis driven by a germline-specific gene (nos-Gal4, UASp-flp) to induce GSC-like cells mutant for bam or bam's co-factor bgcn. Bam-mutant or bgcn-mutant germ cells produce tumors in the stem cell compartment (the germarium) of the ovary (Figure 1). These tumors contain non-mutant cells - termed SGC for single-germ cells. 75% of SGCs do not exhibit signs of differentiation (as assessed by bamP-GFP) (Figure 2). The authors demonstrate that block in differentiation in SGC is a result of suppression of bam expression (Figure 2). They present data suggesting that in 73% of SGCs, BMP signaling is low (assessed by dad-lacZ) (Figure 3) and proliferation is less in SGCs vs GSCs. They present genetic evidence that mutations in BMP pathway receptors and transcription factors suppress some of the non-autonomous effects exhibited by SGCs within bam-mutant tumors (Figure 4). They show data that bam-mutant cells secrete Dpp, but this data is not compelling (see below) (Figure 5). They provide genetic data that loss of BMP ligands (dpp and gbb) suppresses the appearance of SGCs in bam-mutant tumors (Figure 6). Taken together, their data support a model in which bam-mutant GSC-like cells produce BMPs that act on non-mutant cells (i.e., SGCs) to prevent their differentiation, similar to what is seen in the ovarian stem cell niche.
This preprint from Shaowei Zhao and colleagues presents results that suggest tumorous germline stem cells (GSCs) in the Drosophila ovary mimic the ovarian stem cell niche and inhibit the differentiation of neighboring non-mutant GSC-like cells. The authors use FRT-mediated clonal analysis driven by a germline-specific gene (nos-Gal4, UASp-flp) to induce GSC-like cells mutant for bam or bam's co-factor bgcn. Bam-mutant or bgcn-mutant germ cells produce tumors in the stem cell compartment (the germarium) of the ovary (Figure 1). These tumors contain non-mutant cells - termed SGC for single-germ cells. 75% of SGCs do not exhibit signs of differentiation (as assessed by bamP-GFP) (Figure 2). The authors demonstrate that block in differentiation in SGC is a result of suppression of bam expression (Figure 2). They present data suggesting that in 73% of SGCs, BMP signaling is low (assessed by dad-lacZ) (Figure 3) and proliferation is less in SGCs vs GSCs. They present genetic evidence that mutations in BMP pathway receptors and transcription factors suppress some of the non-autonomous effects exhibited by SGCs within bam-mutant tumors (Figure 4). They

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Strengths:

(1) Use of an excellent and established model for tumorous cells in a stem cell microenvironment.

(2) Powerful genetics allow them to test various factors in the tumorous vs non-tumorous cells.

(3) Appropriate use of quantification and statistics.

Weaknesses:

(1) What is the frequency of SGCs in nos>flp; bam-mutant tumors? For example, are they seen in every germarium, or in some germaria, etc, or in a few germaria?

(2) Does the breakdown in clonality vary when they induce hs-flp clones in adults as opposed to in larvae/pupae?

(3) Approximately 20-25% of SGCs are bam+, dad-LacZ+. Firstly, how do the authors explain this? Secondly, of the 70-75% of SGCs that have no/low BMP signaling, the authors should perform additional characterization using markers that are expressed in GSCs (i.e., Sex lethal and nanos).

(4) All experiments except Figure 1I (where a single germarium with no quantification) were performed with nos-Gal4, UASp-flp. Have the authors performed any of the phenotypic characterizations (i.e., figures other than Figure 1) with hs-flp?

(5) Does the number of SGCs change with the age of the female? The experiments were all performed in 14-day-old adult females. What happens when they look at a young female (like 2-day-old). I assume that the nos>flp is working in larval and pupal stages, and so the phenotype should be present in young females. Why did the authors choose this later age? For example, is the phenotype more robust in older females? Or do you see more SGCs at later time points?

(6) Can the authors distinguish one copy of GFP versus 2 copies of GFP in germ cells of the ovary? This is not possible in the Drosophila testis. I ask because this could impact the clonal analyses diagrammed in Figure 4A and 4G and in 6A and B. Additionally, in most of the figures, the GFP is saturated, so it is not possible to discern one vs two copies of GFP.

(7) More evidence is needed to support the claim of elevated Dpp levels in bam or bgcn mutant tumors. The current results with the dpp-lacZ enhancer trap in Figure 5A, B are not convincing. First, why is the dpp-lacZ so much brighter in the mosaic analysis (A) than in the no-clone analysis (B)? It is expected that the level of dpp-lacZ in cap cells should be invariant between ovaries, and yet LacZ is very faint in Figure 5B. I think that if the settings in A matched those in B, the apparent expression of dpp-lacZ in the tumor would be much lower and likely not statistically significant. Second, they should use RNA in situ hybridization with a sensitive technique like hybridization chain reactions (HCR) - an approach that has worked well in numerous Drosophila tissues, including the ovary.

(8) In Figure 6, the authors report results obtained with the bamBG allele. Do they obtain similar data with another bam allele (i.e., bamdelta86)?

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Reviewer #2 (Public review):

While the study by Zhang et al. provides valuable insights into how germline tumors can non-autonomously suppress the differentiation of neighboring wild-type germline stem cells (GSCs), several conceptual and technical issues limit the strength of the conclusions.

Major points:

(1) Naming of SGCs is confusing. In line 68, the authors state that "many wild-type germ cells located outside the niche retained a GSC-like single-germ-cell (SGC) morphology." However, bam or bgcn mutant GSCs are also referred to as "SGCs," which creates confusion when reading the text and interpreting the figures. The authors should clarify the terminology used to distinguish between wild-type SGCs and tumor (bam/bgcn mutant) SGCs, and apply consistent naming throughout the manuscript and figure legends.

a) The same confusion appears in Figure 2. It is unclear whether the analyzed SGCs are wild-type or bam mutant cells. If the SGCs analyzed are Bam mutants, then the lack of Bam expression and failure to differentiate would be expected and not informative. However, if the SGCs are wild-type GSCs located outside the niche, then the observation would suggest that Bam expression is silenced in these wild-type cells, which is a significant finding. The authors should clarify the genotype of the SGCs analyzed in Figure 2C, as this information is not currently provided.

b) In Figures 4B and 4E, the analysis of SGC composition is confusing. In the control germaria (bam mutant mosaic), the authors label GFP^+ SGCs as "wild-type," which makes interpretation unclear. Note, this is completely different from their earlier definition shown in line 68.

c) Additionally, $bam^{+/-}$ GSCs (the first bar in Figure 4E) should appear GFP^+ and Red^+ (i.e., yellow). It would be helpful if the authors could indicate these $bam^{+/-}$ germ cells directly in the image and clarify the corresponding color representation in the main text. In Figure 2A, although a color code is shown, the legend does not explain it clearly, nor does it specify the identity of $bam^{+/-}$ cells alone. Figure 4F has the same issue, and in this graph, the color does not match Figure 4A.

(2) The frequencies of bam or bgcn mutant mosaic germaria carrying [wild-type] SGCs or wild-type germ cell cysts with branched fusomes, as well as the average number of wild-type SGCs per germarium and the number of days after heat shock for the representative images, are not provided when Figure 1 is first introduced. Since this is the first time the authors describe these phenotypes, including these details is essential. Without this information, it is difficult for readers to follow and evaluate the presented observations.

(3) Without the information mentioned in point 2, it causes problems when reading through the section regarding [wild-type] SGCs induced by impairment of differentiation or dedifferentiation. In lines 90-97, the authors use the presence of midbodies between cystocytes as a criterion to determine whether the wild-type GSCs surrounded by tumor GSCs arise through dedifferentiation. However, the cited study (Mathieu et al., 2022) reports that midbodies can be detected between two germ cells within a cyst carrying a branched fusome upon USP8 loss.

a) Are wild-type germ cell cysts with branched fusomes present in the bam mutant mosaic germaria? What is the proportion of germaria containing wild-type SGCs versus those containing wild-type germ cell cysts with branched fusomes?

b) If all bam mutant mosaic germaria carry only wild-type GSCs outside the niche and no germaria contain wild-type germ cell cysts with branched fusomes, then examining midbodies as an indicator of dedifferentiation may not be appropriate.

c) If, however, some germaria do contain wild-type germ cell cysts with branched fusomes, the authors should provide representative images and quantify their proportion.

d) In line 95, although the authors state that 50 germ cell cysts were analyzed for the presence of midbodies, it would be more informative to specify how many germaria these cysts were derived from and how many biological replicates were examined.

(4) Note that both bam mutant GSCs and wild-type SGCs can undergo division to generate midbodies (double cells), as shown in Figure 4H. Therefore, the current description of the midbody analysis is confusing. The authors should clarify which cell types were examined and explain how midbodies were interpreted in distinguishing between cell division and differentiation.

(5) The data in Figure 5 showing Dpp expression in bam mutant tumorous GSCs are not convincing. The Dpp-lacZ signal appears broadly distributed throughout the germarium, including in escort cells. To support the claim more clearly, the authors should present corresponding images for Figures 5D and 5E, in which dpp expression was knocked down in the germ cells of bam or bgcn mutant mosaic germaria. Showing these images would help clarify the localization and specificity of Dpp-lacZ expression relative to the tumorous GSCs.

(6) While Figure 6 provides genetic evidence that bam mutant tumorous GSCs produce Dpp to inhibit the differentiation of wild-type SGCs, it should be noted that these analyses were performed in a $dpp^{+/-}$ background. To strengthen the conclusion, the authors should include appropriate controls showing [$dpp^{+/-}$; $bam^{+/-}$] SGCs and [$dpp^{+/-}$; $bam^{+/-}$] germ cell cysts without heat shock (as referenced in Figures 6F and 6I).

(7) Previous studies have reported that bam mutant germ cells cause blunted escort cell protrusions (e.g., Kirilly et al., Development, 2011), which are known to contribute to germ cell differentiation (e.g., Chen et al., Frontiers in Cell and Developmental Biology, 2022). The authors should include these findings in the Discussion to provide a broader context and to acknowledge how alterations in escort cell morphology may further influence differentiation defects in their model.

(8) Since fusome morphology is an important readout of SGCs vs differentiation. All the clonal analysis should have fusome staining.

(9) Figure arrangement. It is somewhat difficult to identify the figure panels cited in the text due to the current panel arrangement.

(10) The number of biological replicates and germaria analyzed should be clearly stated somewhere in the manuscript-ideally in the Methods section or figure legends. Providing this information is essential for assessing data reliability and reproducibility.

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Reviewer #3 (Public review):

Summary:

Zhang et al. investigated how germline tumors influence the development of neighboring wild-type (WT) germline stem cells (GSC) in the Drosophila ovary. They report that germline tumors inhibit the differentiation of neighboring WT GSCs by arresting them in an undifferentiated state, resulting from reduced expression of the differentiation-promoting factor Bam. They find that these tumor cells produce low levels of the niche-associated signaling molecules Dpp and Gbb, which suppress bam expression and consequently inhibit the differentiation of neighboring WT GSCs non-cell-autonomously. Based on these findings, the authors propose that germline tumors mimic the niche to suppress the differentiation of the neighboring stem cells.

Strengths:

This study addresses an important biological question concerning the interaction between germline tumor cells and WT germline stem cells in the Drosophila ovary. If the findings are substantiated, they could provide valuable insights applicable to other stem cell systems.

Weaknesses:

Previous work from Xie's lab demonstrated that bam and bgcn mutant GSCs can outcompete WT GSCs for niche occupancy. Furthermore, a large body of literature has established that the interactions between escort cells (ECs) and GSC daughters are essential for proper and timely germline differentiation (the differentiation niche). Disruption of these interactions leads to arrest of germline cell differentiation in a status with weak BMP signaling activation and low bam expression, a phenotype virtually identical to what is reported here.

Thus, it remains unclear whether the observed phenotype reflects "direct inhibition by tumor cells" or "arrested differentiation due to the loss of the differentiation niche". Because most data were collected at a very late stage (more than 10 days after clonal induction), when tumor cells already dominate the germarium, this question cannot be solved. To distinguish between these two possibilities, the authors could conduct a time-course analysis to examine the onset of the WT GSC-like single-germ-cell (SGC) phenotype and determine whether early-stage tumor clones with a few tumor cells can suppress the differentiation of neighboring WT GSCs with only a few tumor cells present. If tumor cells indeed produce Dpp and Gbb (as proposed here) to inhibit the differentiation of neighboring germline cells, a small cluster or probably even a single tumor cell generated at an early stage might prevent the differentiation of their neighboring germ cells.

The key evidence supporting the claim that tumor cells produce Gpp and Gbb comes from Figures 5 and 6, which suggest that tumor-derived dpp and gbb are required for this inhibition. However, interpretation of these data requires caution.

In Figure 5, the authors use dpp-lacZ to support the claim that dpp is upregulated in tumor cells (Figure 5A and 5B). However, the background expression in somatic cells (ECs and pre-follicular cells) differs noticeably between these panels. In Figure 5A, dpp-lacZ expression in somatic cells in 5A is clearly higher than in 5B, and the expression level in tumor cells appears comparable to that in somatic cells (dpp-lacZ single channel). Similarly, in Figure 5B, dpp-lacZ expression in germline cells is also comparable to that in somatic cells. Providing clear evidence of upregulated dpp and gbb expression in tumor cells (for example, through single-molecular RNA in situ) would be essential.

Most tumor data present in this study were collected from the bam[86] null allele, whereas the data in Figure 6 were derived from a weaker bam[BG] allele. This bam[BG] allele is not molecularly defined and shows some genetic interaction with dpp mutants. As shown in Figure 6E, removal of dpp from homozygous bam[BG] mutant leads to germline differentiation (evidenced by a branched fusome connecting several cystocytes, located at the right side of the white arrowhead). In Figure 6D, fusome is likely present in some GFP-negative bam[BG]/bam[BG] cells. To strengthen their claim that the tumor produces Dpp and Gbb to inhibit WT germline cell differentiation, the authors should repeat these experiments using the bam[86] null allele.

It is well established that the stem niche provides multiple functional supports for maintaining resident stem cells, including physical anchorage and signaling regulation. In Drosophila, several signaling molecules produced by the niche have been identified, each with a distinct function - some promoting stemness, while others regulate differentiation. Expression of Dpp and Gbb alone does not substantiate the claim that these tumor cells have acquired the niche-like property. To support their assertion that these tumors mimic the niche, the authors should provide additional evidence showing that these tumor cells also express other niche-associated markers. Alternatively, they could revise the manuscript title to more accurately reflect their findings.

In the Method section, the authors need to provide details on how dpp-lacZ expression levels were quantified and normalized.

https://doi.org/10.7554/eLife.108910.1.sa1

Author response:

> ***Reviewer #1 (Public review):***
>
> *Summary:*
>
> *This preprint from Shaowei Zhao and colleagues presents results that suggest tumorous germline stem cells (GSCs) in the Drosophila ovary mimic the ovarian stem cell niche and inhibit the differentiation of neighboring non-mutant GSC-like cells. The authors use FRT-mediated clonal analysis driven by a germline-specific gene (nos-Gal4, UASp-flp) to induce GSC-like cells mutant for bam or bam's cofactor bgcn. Bam-mutant or bgcn-mutant germ cells produce tumors in the stem cell compartment (the germarium) of the ovary (Figure 1). These tumors contain non-mutant cells - termed SGC for single-germ cells. 75% of SGCs do not exhibit signs of differentiation (as assessed by bamP-GFP) (Figure 2). The authors demonstrate that block in differentiation in SGC is a result of suppression of bam expression (Figure 2). They present data suggesting that in 73% of SGCs, BMP signaling is low (assessed by dad-lacZ) (Figure 3) and proliferation is less in SGCs vs GSCs. They present genetic evidence that mutations in BMP pathway receptors and transcription factors suppress some of the non-autonomous effects exhibited by SGCs within bam-mutant tumors (Figure 4). They show data that bam-mutant cells secrete Dpp, but this data is not compelling (see below) (Figure 5). They provide genetic data that loss of BMP ligands (dpp and gbb) suppresses the appearance of SGCs in bam-mutant tumors (Figure 6). Taken together, their data support a model in which bam-mutant GSC-like cells produce BMPs that act on nonmutant cells (i.e., SGCs) to prevent their differentiation, similar to what is seen in the ovarian stem cell niche.*
>
> *Strengths:*
>
> *(1) Use of an excellent and established model for tumorous cells in a stem cell microenvironment.*
>
> *(2) Powerful genetics allow them to test various factors in the tumorous vs nontumorous cells.*
>
> *(3) Appropriate use of quantification and statistics.*

We greatly appreciate these comments.

> *Weaknesses:*

> *(1) What is the frequency of SGCs in nos>flp; bam-mutant tumors? For example, are they seen in every germarium, or in some germaria, etc, or in a few germaria?*

This is a great question. Because the SGC phenotype depends on the presence of germline tumor clones, our quantification was restricted to germaria that contained them.These quantification data ("SGCs and/or germline cysts per germarium with germline clones") will be presented in the revised Figure 1.

> *(2) Does the breakdown in clonality vary when they induce hs-flp clones in adults as opposed to in larvae/pupae?*

Our initial attempts to induce ovarian hs-flp germline clones by heat-shocking adult flies were unsuccessful, with very few clones being observed. Therefore, we shifted our approach to an earlier developmental stage. Successful induction was achieved by subjecting late-L3/early-pupal animals to a twice-daily heatshock at 37°C for 6 consecutive days (2 hours per session with a 6-hour interval, see Lines 325-329) (Zhao et al., 2018).

> *(3) Approximately 20-25% of SGCs are bam+, dad-LacZ+. Firstly, how do the authors explain this? Secondly, of the 70-75% of SGCs that have no/low BMP signaling, the authors should perform additional characterization using markers that are expressed in GSCs (i.e., Sex lethal and nanos).*

These 20-25% of SGCs are bamP-GFP^{+} dad-lacZ-, not bam^{+} dad-lacZ^{+} (see Figure 2C and 3D). They would be cystoblast-like cells that may have initiated a differentiation program toward forming germline cysts (see Lines 109-117). The 70-75% of SGCs that have low BMP signaling exhibit GSC-like properties, including: 1) dot-like spectrosomes; 2) dad-lacZ positivity; 3) absence of bamP-GFP expression. While additional markers would be beneficial, we think that this combination of properties is sufficient to classify these cells as GSC-like.

> *(4) All experiments except Figure 1I (where a single germarium with no quantification) were performed with nos-Gal4, UASp-flp. Have the authors performed any of the phenotypic characterizations (i.e., figures other than Figure 1) with hs-flp?*

Yes, we initially identified the SGC phenotype through hs-flp-mediated mosaic analysis of bam or bgcn mutant in ovaries. However, as noted in our response to Weakness (2), this approach was very labor-intensive. Therefore, we switched to using the more convenient nos::flp system for subsequent experiments. To our observation, there was no difference in the SGC phenotype between these two approaches, confirming that the nos::flp system is a valid and more practical alternative for its study.

> *(5) Does the number of SGCs change with the age of the female? The experiments were all performed in 14-day-old adult females. What happens when they look at a young female (like 2-day-old). I assume that the nos>flp is working in larval and pupal stages, and so the phenotype should be present in young females. Why did the authors choose this later age? For example, is the phenotype more robust in older females? Or do you see more SGCs at later time points?*

These are very good questions. Such time-course analysis data will be provided in revised Figure 1. The SGC phenotype depends on the presence of bam or bgcn mutant germline clones. Germaria from 14-day-old flies contained bigger and more such clones than those from younger flies. This age-dependent increase in clone size and frequency significantly enhanced the efficiency of our quantification (see Lines 129-131).

> *(6) Can the authors distinguish one copy of GFP versus 2 copies of GFP in germ cells of the ovary? This is not possible in the Drosophila testis. I ask because this could impact the clonal analyses diagrammed in Figure 4A and 4G and in 6A and B. Additionally, in most of the figures, the GFP is saturated, so it is not possible to discern one vs two copies of GFP.*

We greatly appreciate this comment. It was also difficult for us to distinguish 1 and 2 copies of GFP in the Drosophila ovary. In Figure 4A-F, to resolve this problem, we used a triplecolor system, in which red germ cells ($RFP^{+/+}$ $GFP^{-/-}$) are bam mutant, yellow germ cells ($RFP^{+/-}$ $GFP^{+/-}$) are wild-type, and green germ cells ($RFP^{-/-}$ $GFP^{+/+}$) are punt or med mutant. In Figure 4G-J, we quantified the SGC phenotype only in black germ cells ($GFP^{-/-}$), which are wild-type (control) or mad mutant. In Figure 6, we quantified the SGC phenotype only in green germ cells (both $GFP^{+/+}$ and $GFP^{+/-}$), all of which are wild-type.

> *(7) More evidence is needed to support the claim of elevated Dpp levels in bam or bgcn mutant tumors. The current results with the dpp-lacZ enhancer trap in Figure 5A, B are not convincing. First, why is the dpp-lacZ so much brighter in the mosaic analysis (A) than in the no-clone analysis (B)? It is expected that the level of dpplacZ in cap cells should be invariant between ovaries, and yet LacZ is very faint in Figure 5B. I think that if the settings in A matched those in B, the apparent expression of dpp-lacZ in the tumor would be much lower and likely not statistically significant. Second, they should use RNA in situ hybridization with a sensitive technique like hybridization chain reactions (HCR) - an approach that has worked well in numerous Drosophila tissues, including the ovary.*

We appreciate this critical comment. The settings of immunofluorescent staining and confocal parameters in Figure 5A were the same as those in 5B. To our observation, the level of dpp-lacZ in cap cells was variable across germaria, even within the same ovary, as quantified in Figure 5C. We will provide RNA in situ hybridization data to further strengthen the conclusion that bam or bgcn mutant germline tumors secret BMP ligands.

> *(8) In Figure 6, the authors report results obtained with the bamBG allele. Do they obtain similar data with another bam allele (i.e., bamdelta86)?*

No. Given that bam^{BG} was functionally indistinguishable from $bam^{\Delta 86}$ in inducing the SGC phenotype (compare Figure 6F, I with Figure 6-figure supplement 3C), we believe that repeating these experiments with $bam^{\Delta 86}$ would be redundant and would not alter the key conclusion of our study. Thanks for the understanding!

> ***Reviewer #2 (Public review):***
>
> *While the study by Zhang et al. provides valuable insights into how germline tumors can non-autonomously suppress the differentiation of neighboring wild-type germline stem cells (GSCs), several conceptual and technical issues limit the strength of the conclusions.*
>
> *Major points:*
>
> *(1) Naming of SGCs is confusing. In line 68, the authors state that "many wild-type germ cells located outside the niche retained a GSC-like single-germ-cell (SGC) morphology." However, bam or bgcn mutant GSCs are also referred to as "SGCs," which creates confusion when reading the text and interpreting the figures. The authors should clarify the terminology used to distinguish between wild-type SGCs and tumor (bam/bgcn mutant) SGCs, and apply consistent naming throughout the manuscript and figure legends.*

We apologize for any confusion. In our manuscript, the term "SGC" is reserved specifically for wild-type germ cells that maintain a GSC-like morphology outside the niche. bam or bgcn mutant germ cells are referred to as GSC-like tumor cells (Lines 87-88), not SGCs.

> *(a) The same confusion appears in Figure 2. It is unclear whether the analyzed SGCs are wild-type or bam mutant cells. If the SGCs analyzed are Bam mutants, then the lack of Bam expression and failure to differentiate would be expected and not informative. However, if the SGCs are wild-type GSCs located outside the niche, then the observation would suggest that Bam expression is silenced in these wildtype cells, which is a significant finding. The authors should clarify the genotype of the SGCs analyzed in Figure 2C, as this information is not currently provided.*

The SGCs analyzed in Figure 2A-C are wild-type, GSC-like cells located outside the niche. They were generated using the same genetic strategy depicted in Figures 1C and 1E (with the schematic in Figure 1B). The complete genotypes for all experiments are available in

Source data 1.

> *(b) In Figures 4B and 4E, the analysis of SGC composition is confusing. In the control germaria (bam mutant mosaic), the authors label GFP^{+} SGCs as "wild-type," which makes interpretation unclear. Note, this is completely different from their earlier definition shown in line 68.*

The strategy to generate SGCs in Figure 4B-F (with the schematic in Figure 4A) is completely different from that in Figure 1C-F, H, and I (with the schematic in Figure 1B). In Figure 4B-F, we needed to distinguish punt$^{-/-}$ (or med$^{-/-}$) with punt$^{+/-}$ (or med$^{+/-}$) germ cells. As noted in our response to Reviewer #1's Weakness (6), it was difficult for us to distinguish 1 and 2 copies of GFP in the Drosophila ovary. Therefore, we chose to use the triple-color system to distinguish these germ cells in Figure 4B-F (see genotypes in Source data 1).

> *(c) Additionally, bam$^{+/-}$ GSCs (the first bar in Figure 4E) should appear GFP^{+} and Red^{+} (i.e., yellow). It would be helpful if the authors could indicate these bam$^{+/-}$ germ cells directly in the image and clarify the corresponding color representation in the main text. In Figure 2A, although a color code is shown, the legend does not explain it clearly, nor does it specify the identity of bam$^{+/-}$ cells alone. Figure 4F has the same issue, and in this graph, the color does not match Figure 4A.*

The color-to-genotype relationships for the schematics in Figures 2A and 4E are provided in Figures 1B and 4A, respectively. Due to the high density of germ cells, it is impractical to label each genotype directly in the images. In contrast to Figure 4E, the colors in Figure 4F do not represent genotypes; instead, blue denotes the percentage of SGCs, and red denotes the percentage of germline cysts, as indicated below the bar chart.

> *(2) The frequencies of bam or bgcn mutant mosaic germaria carrying [wild-type] SGCs or wild-type germ cell cysts with branched fusomes, as well as the average number of wild-type SGCs per germarium and the number of days after heat shock for the representative images, are not provided when Figure 1 is first introduced. Since this is the first time the authors describe these phenotypes, including these details is essential. Without this information, it is difficult for readers to follow and evaluate the presented observations.*

Thanks for this constructive suggestion. We will include such quantification data in the revised manuscript.

> *(3) Without the information mentioned in point 2, it causes problems when reading through the section regarding [wild-type] SGCs induced by impairment of differentiation or dedifferentiation. In lines 90-97, the authors use the presence of midbodies between cystocytes as a criterion to determine whether the wild-type GSCs surrounded by tumor GSCs arise through dedifferentiation. However, the cited study (Mathieu et al., 2022) reports that midbodies can be detected between two germ cells within a cyst carrying a branched fusome upon USP8 loss.*

Unlike wild-type cystocytes, which undergo incomplete cytokinesis and lack midbodies, those with USP8 loss undergo complete cell division, with the presence of midbodies (white arrow, Figure 1F' from Mathieu et al., 2022) as a marker of the late cytokinesis stage (Mathieu et al., 2022).

> *(a) Are wild-type germ cell cysts with branched fusomes present in the bam mutant mosaic germaria? What is the proportion of germaria containing wild-type SGCs versus those containing wild-type germ cell cysts with branched fusomes?*
>
> *(b) If all bam mutant mosaic germaria carry only wild-type GSCs outside the niche and no germaria contain wild-type germ cell cysts with branched fusomes, then examining midbodies as an indicator of dedifferentiation may not be appropriate.*

We greatly appreciate this critical comment. bam mutant mosaic germaria indeed contained wild-type germline cysts, as evidenced by an SGC frequency of ~70%, rather than 100% (see Figures 2H, 4F, 4J, 6F, 6I, and Figure 6-figure supplement 3C). Since the SGC phenotype depends on the presence of bam or bgcn mutant germline tumors, we quantified it as "the percentage of SGCs relative to the total number of SGCs and germline cysts that are surrounded by germline tumors" (see Lines 124-129). Quantifying the SGC phenotype as "the percentage of germaria with SGCs" would be imprecise. This is because the presence and number of SGCs were highly variable among germaria with bam mutant germline clones, and a small number of germaria entirely lacked these clones. We will provide the data of "SGCs and/or germline cysts per germarium with germline clones" in revised Figure 1.

> *(c) If, however, some germaria do contain wild-type germ cell cysts with branched fusomes, the authors should provide representative images and quantify their proportion.*

Such representative germaria are shown in Figure 2G, 3B, 3C, 6D, 6E, and 6H. The percentage of germline cysts can be calculated by "100% - SGC%".

> *(d) In line 95, although the authors state that 50 germ cell cysts were analyzed for the presence of midbodies, it would be more informative to specify how many germaria these cysts were derived from and how many biological replicates were examined.*

As noted in our response to points a) and b) above, the germ cells surrounded by germline tumors, rather than germarial numbers, are more precise for analyzing the phenotype. For this experiment, we examined >50 such germline cysts via confocal microscopy. As the analysis was performed on a defined cellular population, this sample size should be sufficient to support our conclusion.

> *(4) Note that both bam mutant GSCs and wild-type SGCs can undergo division to generate midbodies (double cells), as shown in Figure 4H. Therefore, the current description of the midbody analysis is confusing. The authors should clarify which cell*

> *types were examined and explain how midbodies were interpreted in distinguishing between cell division and differentiation.*

We assayed for the presence of midbodies or not specifically within the germline cysts surrounded by bam mutant tumors, not within the tumors themselves (Lines 94-95). As detailed in Lines 88-97, the absence of midbodies was used as a key criterion to exclude the possibility of dedifferentiation.

> *(5) The data in Figure 5 showing Dpp expression in bam mutant tumorous GSCs are not convincing. The Dpp-lacZ signal appears broadly distributed throughout the germarium, including in escort cells. To support the claim more clearly, the authors should present corresponding images for Figures 5D and 5E, in which dpp expression was knocked down in the germ cells of bam or bgcn mutant mosaic germaria. Showing these images would help clarify the localization and specificity of Dpp-lacZ expression relative to the tumorous GSCs.*

We greatly appreciate this comment. RNA in situ hybridization data will be provided to further strengthen the conclusion that bam or bgcn mutant germline tumors secret BMP ligands.

> *(6) While Figure 6 provides genetic evidence that bam mutant tumorous GSCs produce Dpp to inhibit the differentiation of wild-type SGCs, it should be noted that these analyses were performed in a dpp $^{+/-}$ background. To strengthen the conclusion, the authors should include appropriate controls showing [dpp $^{+/-}$; bam $^{+/-}$] SGCs and [dpp $^{+/-}$; bam $^{+/-}$] germ cell cysts without heat shock (as referenced in Figures 6F and 6I).*

Schematic cartoons in Figure 6A and 6B demonstrate that these analyses were performed in a dpp$^{+/-}$ background. Figure 6-figure supplement 1 indicates that dpp$^{+/-}$ or gbb$^{+/-}$ does not affect GSC maintenance, germ cell differentiation, and female fly fertility. Figure 6C is the control for 6D and 6E, and 6G is the control for 6H, with quantification in 6F and 6I. We used nos::flp, not the heat shock method, to induce germline clones in these experiments (see genotypes in Source data 1).

> *(7) Previous studies have reported that bam mutant germ cells cause blunted escort cell protrusions (e.g., Kirilly et al., Development, 2011), which are known to contribute to germ cell differentiation (e.g., Chen et al., Frontiers in Cell and Developmental Biology, 2022). The authors should include these findings in the Discussion to provide a broader context and to acknowledge how alterations in escort cell morphology may further influence differentiation defects in their model.*

Thanks for teaching us! Such discussion will be included in the revised manuscript.

> *(8) Since fusome morphology is an important readout of SGCs vs differentiation. All the clonal analysis should have fusome staining.*

SGC is readily distinguishable from multi-cellular germline cyst based on morphology. In some clonal analysis experiments, fusome staining was not feasible due to technical limitations such as channel saturation or antibody incompatibility. Thanks for the understanding!

> *(9) Figure arrangement. It is somewhat difficult to identify the figure panels cited in the text due to the current panel arrangement.*

The figure panels were arranged to optimize space while ensuring that related panels are grouped in close proximity for logical comparison. We would be happy to consider any specific suggestions for an alternative layout that could improve clarity. Thanks!

> *(10) The number of biological replicates and germaria analyzed should be clearly stated somewhere in the manuscript-ideally in the Methods section or figure legends. Providing this information is essential for assessing data reliability and reproducibility.*

Thanks for this constructive suggestion. Such information will be included in figure legends in the revised manuscript.

> ***Reviewer #3 (Public review):***
>
> *Summary:*
>
> *Zhang et al. investigated how germline tumors influence the development of neighboring wild-type (WT) germline stem cells (GSC) in the Drosophila ovary. They report that germline tumors inhibit the differentiation of neighboring WT GSCs by arresting them in an undifferentiated state, resulting from reduced expression of the differentiation-promoting factor Bam. They find that these tumor cells produce low levels of the niche-associated signaling molecules Dpp and Gbb, which suppress bam expression and consequently inhibit the differentiation of neighboring WT GSCs non-cell-autonomously. Based on these findings, the authors propose that germline tumors mimic the niche to suppress the differentiation of the neighboring stem cells.*
>
> *Strengths:*
>
> *This study addresses an important biological question concerning the interaction between germline tumor cells and WT germline stem cells in the Drosophila ovary. If the findings are substantiated, they could provide valuable insights applicable to other stem cell systems.*

We greatly appreciate these comments.

> *Weaknesses:*
>
> *Previous work from Xie's lab demonstrated that bam and bgcn mutant GSCs can outcompete WT GSCs for niche occupancy. Furthermore, a large body of literature has established that the interactions between escort cells (ECs) and GSC daughters are essential for proper and timely germline differentiation (the differentiation niche). Disruption of these interactions leads to arrest of germline cell differentiation in a status with weak BMP signaling activation and low bam expression, a phenotype virtually identical to what is reported here. Thus, it remains unclear whether the observed phenotype reflects "direct inhibition by tumor cells" or "arrested differentiation due to the loss of the differentiation niche". Because most data were collected at a very late stage (more than 10 days after clonal induction), when tumor cells already dominate the germarium, this question cannot be solved. To distinguish between these two possibilities, the authors could conduct a time-course analysis to examine the onset of the WT GSC-like singlegerm-cell (SGC) phenotype and determine whether early-stage tumor clones with a few tumor cells can suppress the differentiation of neighboring WT GSCs with only a few tumor cells present. If tumor cells indeed produce Dpp and Gbb (as proposed here) to inhibit the differentiation of neighboring germline cells, a small cluster or probably even a single tumor cell generated at an early stage might prevent the differentiation of their neighboring germ cells.*

Thanks for this critical comment. Such time-course analysis data will be provided in revised Figure 1.

> *The key evidence supporting the claim that tumor cells produce Gpp and Gbb comes from Figures 5 and 6, which suggest that tumor-derived dpp and gbb are required for this inhibition. However, interpretation of these data requires caution. In Figure 5, the authors use dpp-lacZ to support the claim that dpp is upregulated in tumor cells (Figure 5A and 5B). However, the background expression in somatic cells (ECs and pre-follicular cells) differs noticeably between these panels. In Figure 5A, dpp-lacZ expression in somatic cells in 5A is clearly higher than in 5B, and the expression level in tumor cells appears comparable to that in somatic cells (dpplacZ single channel). Similarly, in Figure 5B, dpp-lacZ expression in germline cells is also comparable to that in somatic cells. Providing clear evidence of upregulated dpp and gbb expression in tumor cells (for example, through single-molecular RNA in situ) would be essential.*

We greatly appreciate this critical comment. In our data, the expression of dpp-lacZ in cap cells was variable across germaria, even within the same ovary, as quantified in Figure 5C. The images in Figures 5A and 5B were selected as representative examples of positive signaling. To directly address the reviewer's point and strengthen our conclusion, we will perform RNA in situ hybridization data in the revised manuscript to visualize the expression of BMP ligands within the bam or bgcn mutant germline tumor cells.

> *Most tumor data present in this study were collected from the bam[86] null allele, whereas the data in Figure 6 were derived from a weaker bam[BG] allele. This bam[BG] allele is not molecularly defined and shows some genetic interaction with dpp mutants. As shown in Figure 6E, removal of dpp from homozygous bam[BG] mutant leads to germline differentiation (evidenced by a branched fusome connecting several cystocytes, located at the right side of the white arrowhead). In Figure 6D, fusome is likely present in some GFP-negative bam[BG]/bam[BG] cells. To strengthen their claim that the tumor produces Dpp and Gbb to inhibit WT germline cell differentiation, the authors should repeat these experiments using the bam[86] null allele.*

Although a structure resembling a "branched fusome" is visible in Figure 6E (right of the white arrowhead), it is an artifact resulting from the cytoplasm of GFP-positive follicle cells, which also stain for α-Spectrin, projecting between germ cells of different clones (see the merged image). In both our previous (Zhang et al., 2023) and current studies, bam^{BG} was functionally indistinguishable from $bam^{\Delta 86}$ in its ability to block GSC differentiation and induce the SGC phenotype (compare Figure 6F, I with Figure 6-figure supplement 3C). Given this, we believe that repeating the extensive experiments in Figure 6 with the $bam^{\Delta 86}$ allele would be scientifically redundant and would not change the key conclusion of our study. We thank the reviewer for their consideration.

> *It is well established that the stem niche provides multiple functional supports for maintaining resident stem cells, including physical anchorage and signaling regulation. In Drosophila, several signaling molecules produced by the niche have been identified, each with a distinct function - some promoting stemness, while others regulate differentiation. Expression of Dpp and Gbb alone does not substantiate the claim that these tumor cells have acquired the niche-like property. To support their assertion that these tumors mimic the niche, the authors should provide additional evidence showing that these tumor cells also express other niche-associated markers. Alternatively, they could revise the manuscript title to more accurately reflect their findings.*

Dpp and Gbb are the key niche signals from cap cells for maintaining GSC stemness. Our work demonstrates that germline tumors can specifically mimic this signaling function, not the full suite of cap cell properties, to create a non-cell-autonomous differentiation block. The current title "Tumors mimic the niche to inhibit neighboring stem cell differentiation" reflects this precise concept: a partial, functional mimicry of the niche's most relevant activity in this context. We feel it is an appropriate and compelling summary of our main conclusion.

> *In the Method section, the authors need to provide details on how dpp-lacZ expression levels were quantified and normalized.*

Thanks for this suggestion. Such information will be included in the revised manuscript.

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